

CS-3002 Information Security

Assignment 0

Virtualization, Linux Fundamentals & Command-Line Security Tools

Maximum Marks	150 (Part A: 10 Part B: 140)
Deadline	September 8, 2026 , 11:59 PM
Submission Naming	RollNumber-FullName-A0.pdf
Submission Portal	Google Classroom (GCR)

General Instructions

1. Submit the final PDF version of your report using the naming convention RollNumber-FullName-A0.pdf.
2. Files submitted with an incorrect naming convention or file type will result in a marks deduction.
3. You may use ChatGPT or other AI tools for ideas or research, but do not copy content verbatim. Include screenshots of every prompt you used in an Appendix at the end of your report.
4. Your final draft must be submitted on Google Classroom (GCR) before the deadline. Late submissions will not be accepted under any circumstances.
5. Every screenshot included in the report must capture the full screen, including the taskbar and status bar. Cropped or partial screenshots will result in a marks deduction.
6. Read this entire document carefully before starting. The graded assessment tasks begin in Part B, after the environment setup instructions.
7. If Linux is already installed on your machine, you may skip Part A, simply attach one screenshot of your existing Linux desktop in its place.
8. For every task in Part B, record the exact commands you executed (in order) in your report and attach a full-screen terminal screenshot as evidence. This applies to all tasks and will not be repeated under each one.
9. Use the --help flag after any command (e.g., tr --help) to view its available options.
10. Cases of plagiarism or copying will be treated as serious academic violations and handled per university policy.

Part A: Pre-Environment Setup (10 Marks)

This part introduces virtualization concepts and walks you through installing VMware Workstation Pro and Kali Linux. If you already have a Linux environment set up, you may skip directly to Part B after attaching one screenshot of your desktop.

Learning Objectives

- Understand the concept of virtualization
- Install and configure VMware Workstation
- Install Kali Linux inside a virtual machine
- Get familiar with basic Linux terminal navigation

What Is a Virtual Machine?

A virtual machine (VM) is best thought of as a “computer made of software.” It behaves like a physical machine running its own operating system, storage, networking, and applications, while remaining fully isolated from other VMs on the same host. Virtualization achieves this by abstracting an application, guest OS, or storage layer away from the underlying physical hardware. A hypervisor allocates physical resources to each virtual environment, and may run on top of a host OS or directly on the hardware.

Why Virtualization Matters

- Lower costs: fewer physical servers are needed, reducing hardware and maintenance expenses.
- Easier disaster recovery: snapshots allow near-instant backup and recovery, even migrating a VM to a new location within minutes.
- Easier testing: a failed test can simply revert to a prior snapshot instead of starting over.
- Quicker backups: both the virtual server and VM can be backed up and redeployed efficiently.
- Improved productivity: tasks that take days in a physical environment can be completed in minutes, freeing staff for higher-value work.

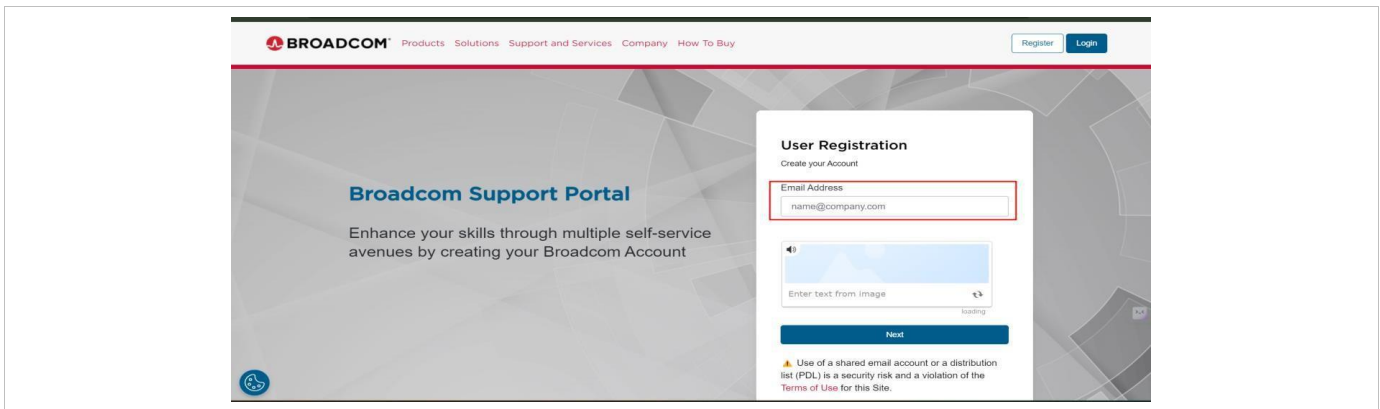
VMware Workstation is one of the most widely used virtualization platforms in the industry, valued for its speed, reliability, scalability, and security. Since most organizations rely on it for internal infrastructure, familiarity with VMware is an essential skill for security professionals.

Tools & Software Required

VMware Workstation Pro	support.broadcom.com/group/ecx/productdownloads?subfamily=VMware Workstation Pro
Kali Linux ISO	cdimage.kali.org/kali-2024.2/kali-linux-2024.2-installer-amd64.iso

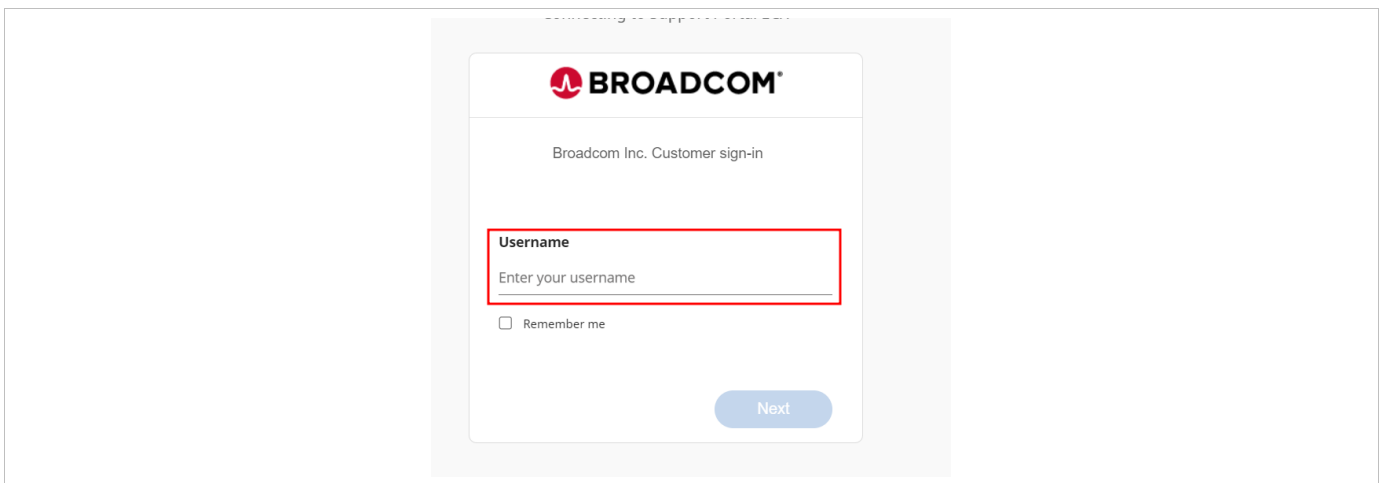
Step 1: Install VMware Workstation Pro

1. Go to profile.broadcom.com/web/registration and register using your university (NU) email address.



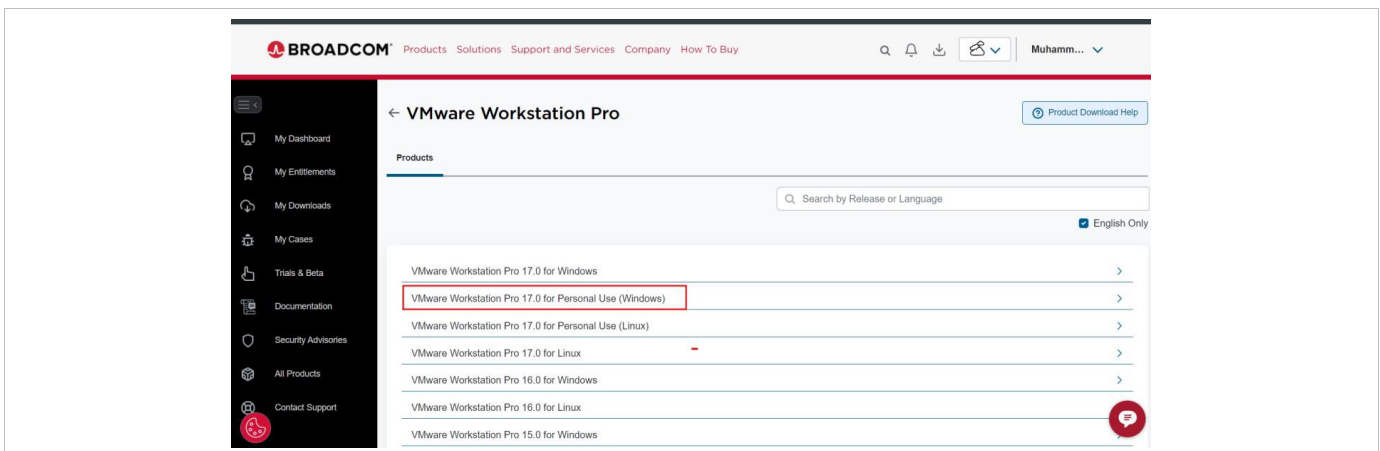
Broadcom Support Portal — account registration

2. Sign in at access.broadcom.com/default/ui/v1/signin/ (you may already be redirected here automatically).



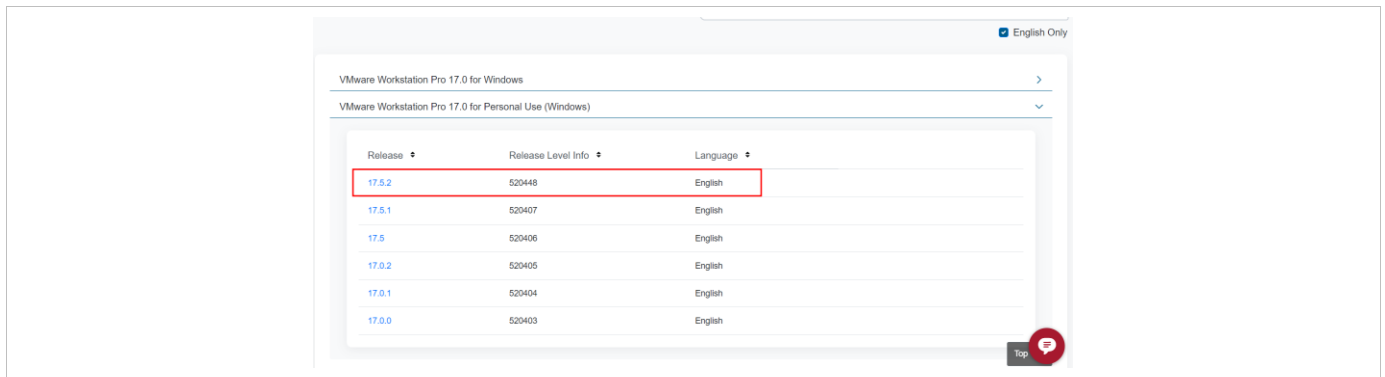
Broadcom Inc. customer sign-in

3. Navigate to the VMware Workstation Pro download page and select the Windows Personal Use tab.



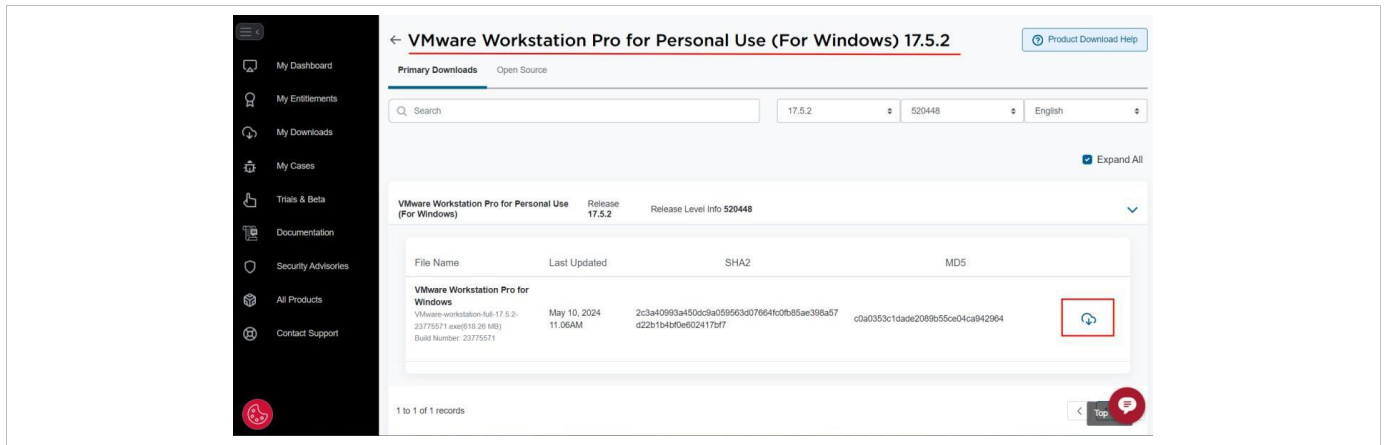
VMware Workstation Pro — product download options

4. Select the latest available release (e.g., 17.5.2).



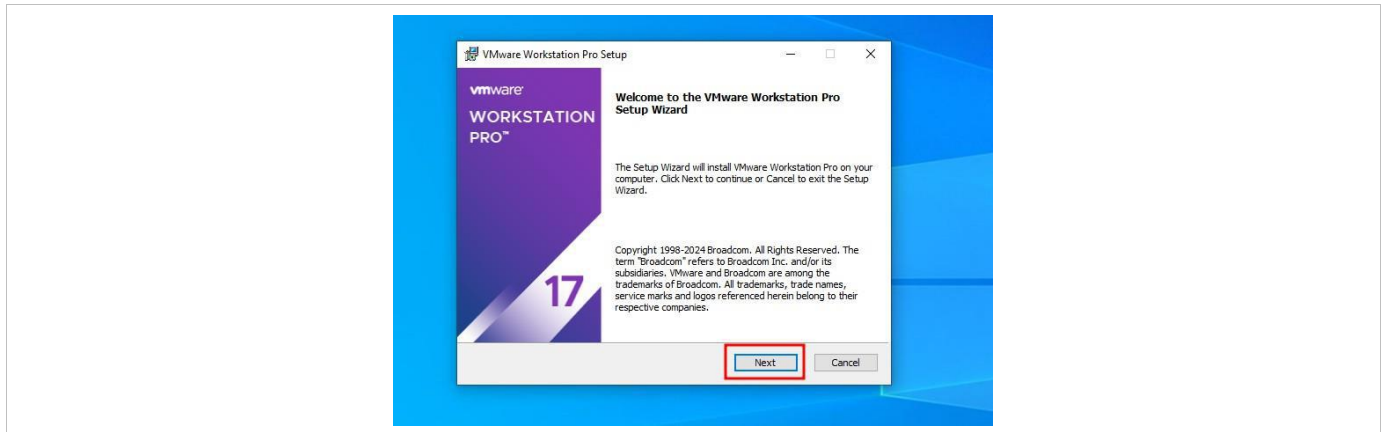
Choosing the latest VMware Workstation Pro release

5. Click the download icon to download the installer.



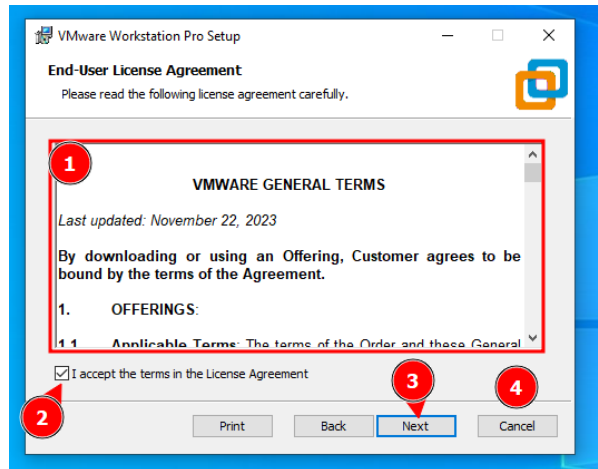
Downloading the VMware Workstation Pro installer

6. Run the downloaded installer and click Next through the welcome screen.



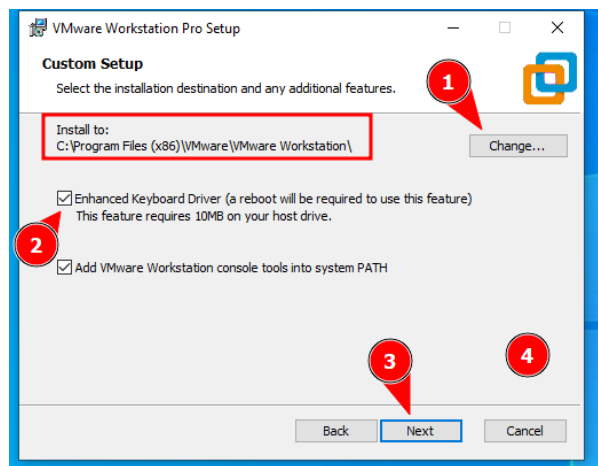
VMware Workstation Pro setup wizard; welcome screen

7. Read and accept the End-User License Agreement, then click Next.



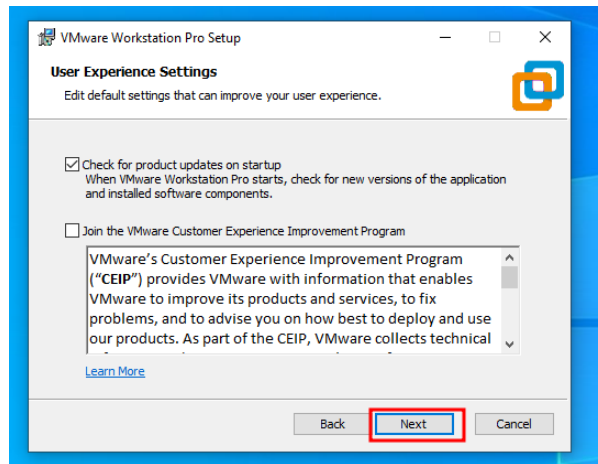
Accepting the End-User License Agreement

8. Confirm or change the installation path, enable Enhanced Keyboard Driver, then click Next.



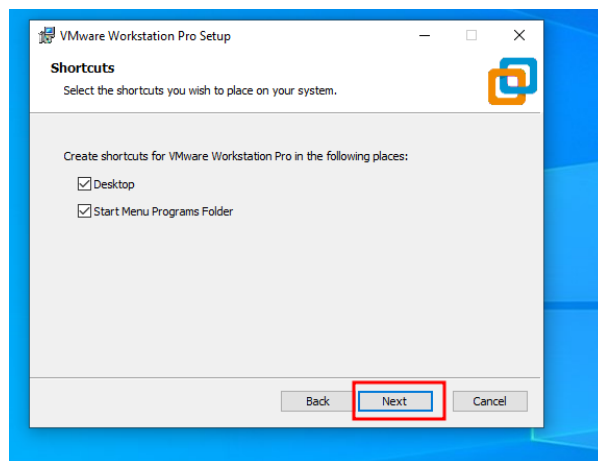
Custom Setup — installation path and features

9. Accept the default user-experience settings and click Next.



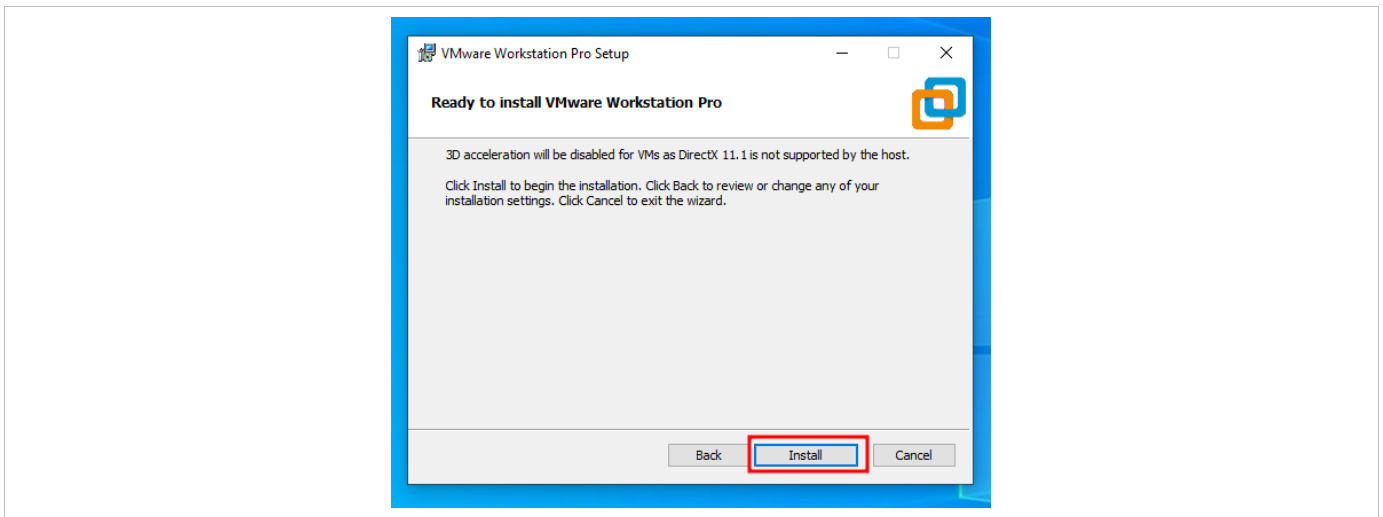
User Experience Settings

10. Choose your preferred desktop/Start Menu shortcuts, then click Next.



Shortcut placement options

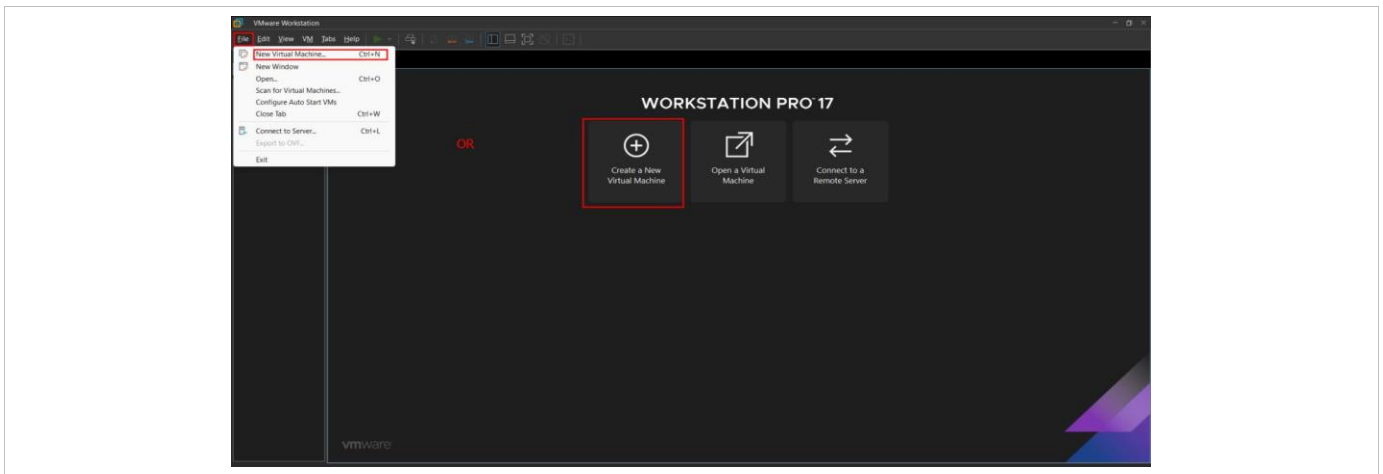
11. Click Install, then Finish once setup completes. VMware Workstation Pro is now installed and can be launched from the desktop shortcut.



Ready to Install VMware Workstation Pro

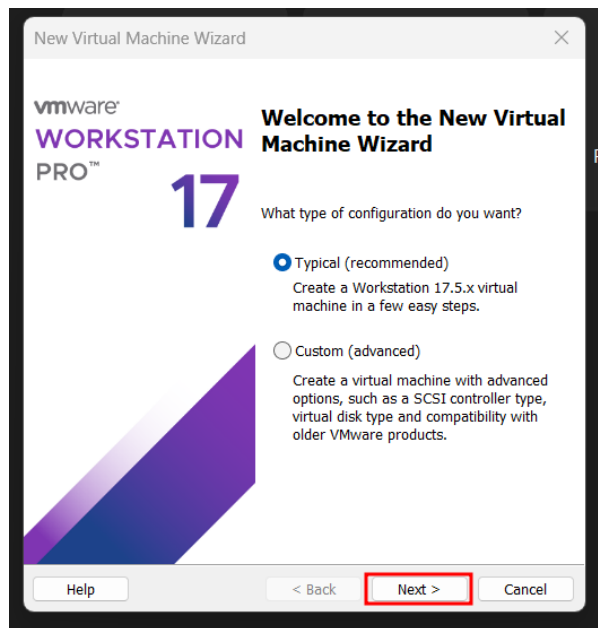
Step 2: Install Kali Linux on VMware Workstation Pro

12. Download the Kali Linux ISO from the link listed in the Tools & Software table above.
13. Open VMware Workstation Pro and select File > New Virtual Machine (or Create a New Virtual Machine from the home screen).



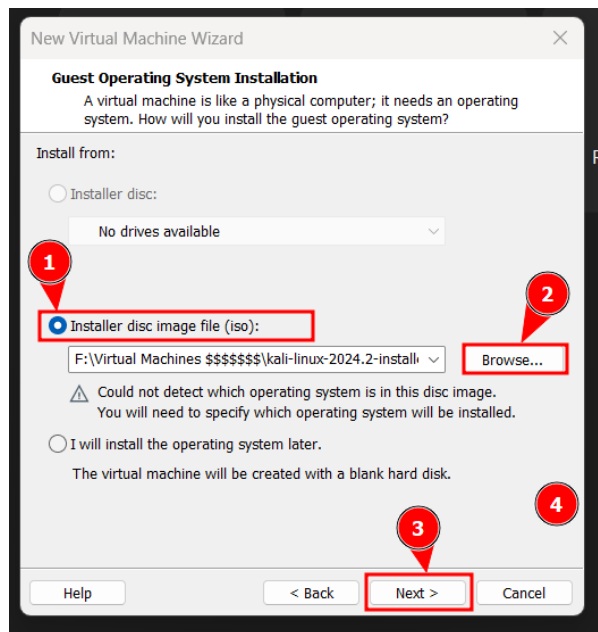
Launching the New Virtual Machine wizard

14. Choose Typical (recommended) configuration and click Next.



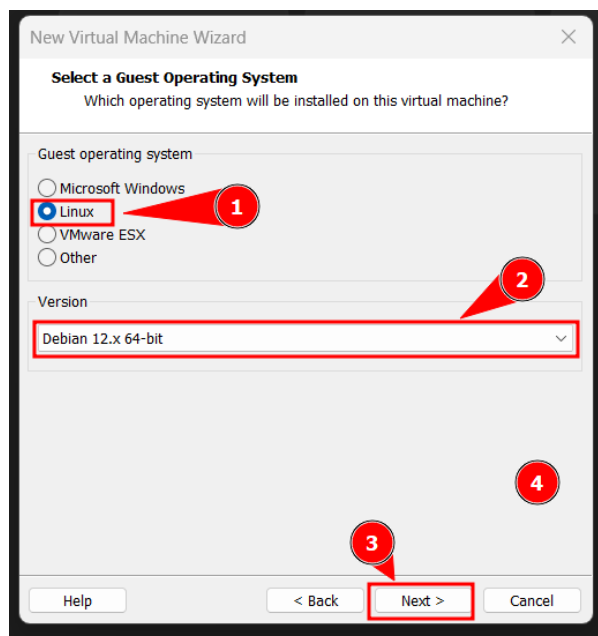
New Virtual Machine Wizard — configuration type

15. Select Installer disc image file (ISO), browse to the downloaded Kali ISO, and click Next.



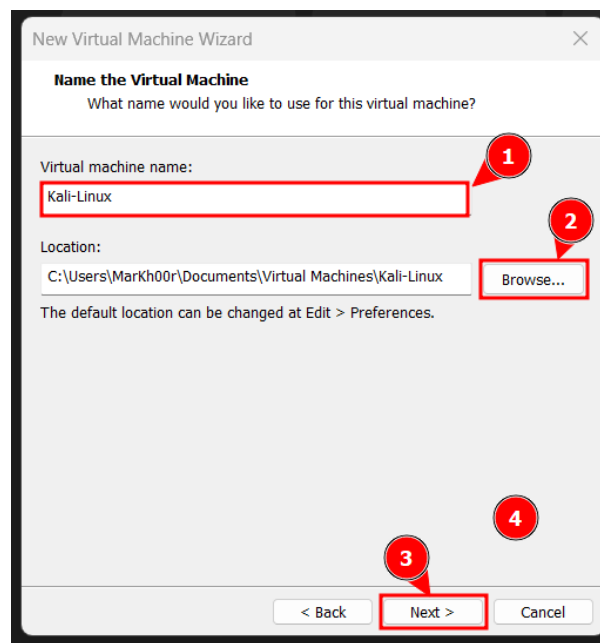
Selecting the Kali Linux ISO as the installer disc image

16. Under Guest Operating System, select Linux, then choose Debian 12.x 64-bit from the version dropdown, and click Next.



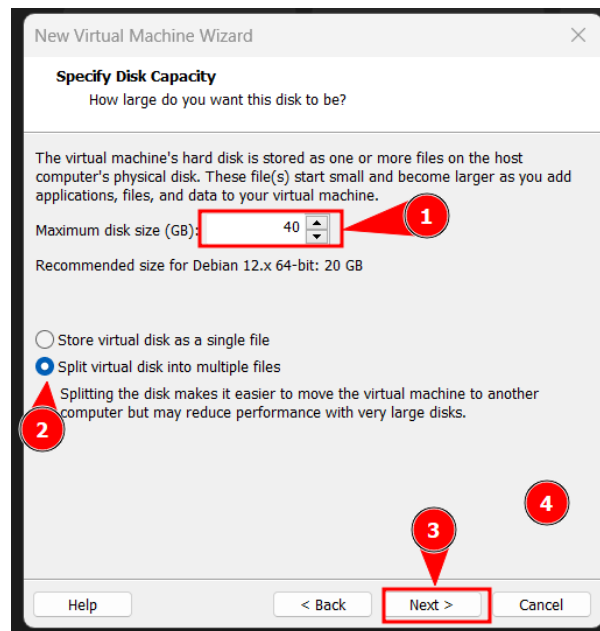
Selecting the guest operating system (Debian 12.x 64-bit)

17. Name the virtual machine (e.g., Kali-Linux) and choose a save location with at least 40 GB of free space, then click Next.



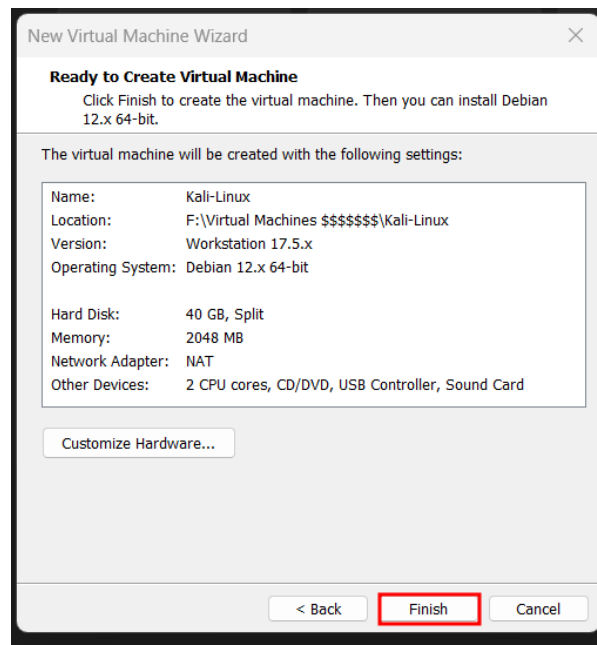
Naming the virtual machine and choosing a save location

18. Set the maximum disk size to at least 40 GB, choose Split virtual disk into multiple files, and click Next.



Specifying disk capacity

19. Review the summary and click Finish to create the VM.



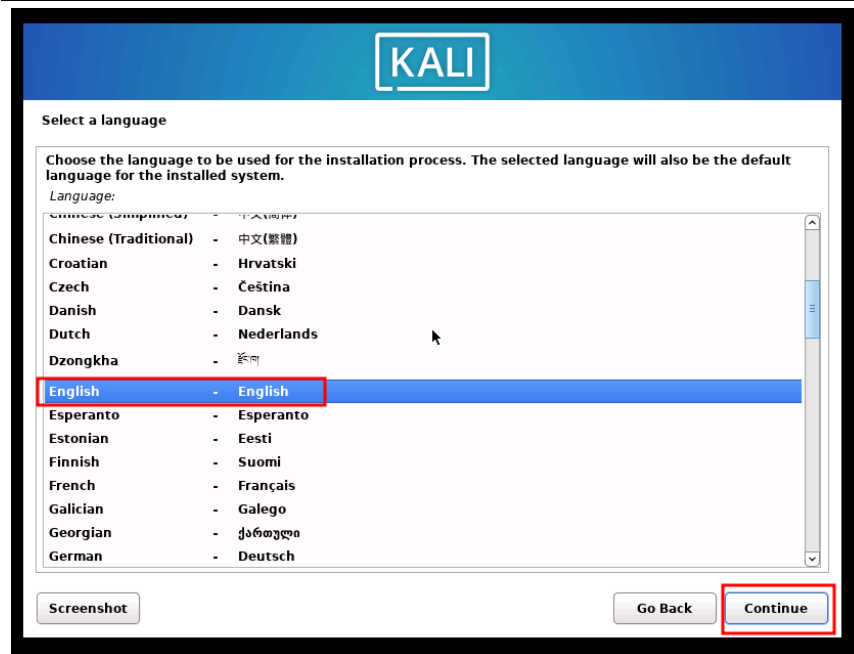
Ready to Create Virtual Machine — summary

20. Power on the new Kali Linux VM and select Graphical Install from the boot menu.



Kali Linux installer boot menu

21. Work through the installer: select your language, location, and keyboard layout; accept or set a hostname; leave the domain name blank; enter your full name and choose a username and password.



KALI

Select your location

The selected location will be used to set your time zone and also for example to help select the system locale. Normally this should be the country where you live.

This is a shortlist of locations based on the language you selected. Choose "other" if your location is not listed.

Country, territory or area:

India

Ireland

Israel

New Zealand

Nigeria

Philippines

Seychelles

Singapore

South Africa

United Kingdom

United States

Zambia

Zimbabwe

other

Screenshot

Go Back

Continue

KALI

Configure the keyboard

Keymap to use:

American English

Albanian

Arabic

Asturian

Bangladesh

Belarusian

Bengali

Belgian

Berber (Latin)

Bosnian

Brazilian

British English

Bulgarian (BDS layout)

Bulgarian (phonetic layout)

Burmese

Canadian French

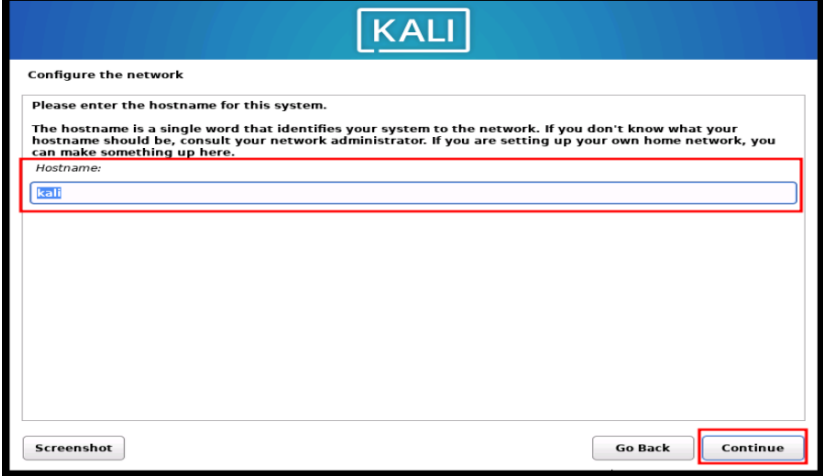
Screenshot

Go Back

Continue

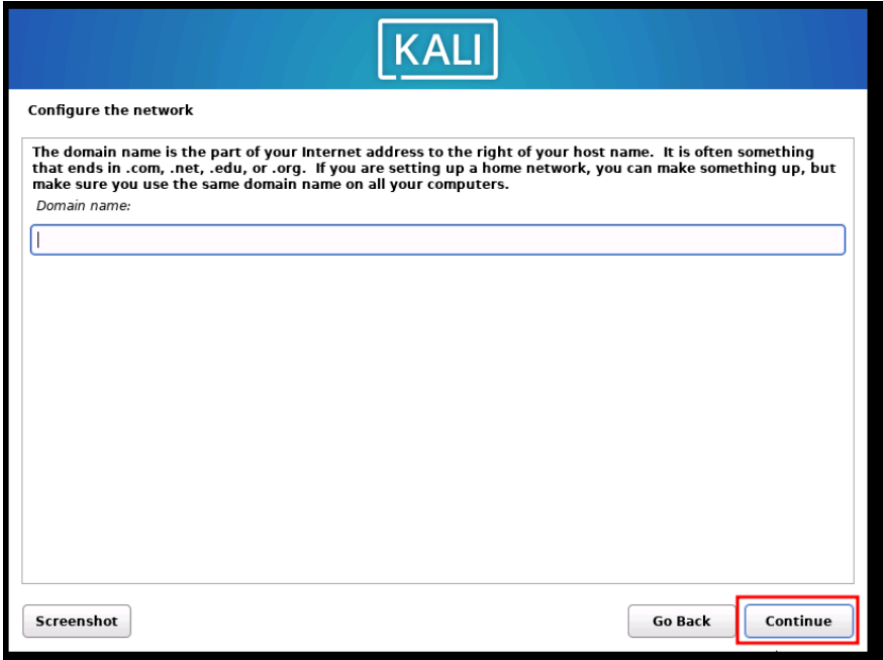
Now wait for the progress; it will take some time to add additional components.

22. Write Hostname or click to continue to use the default hostname 'kali'.



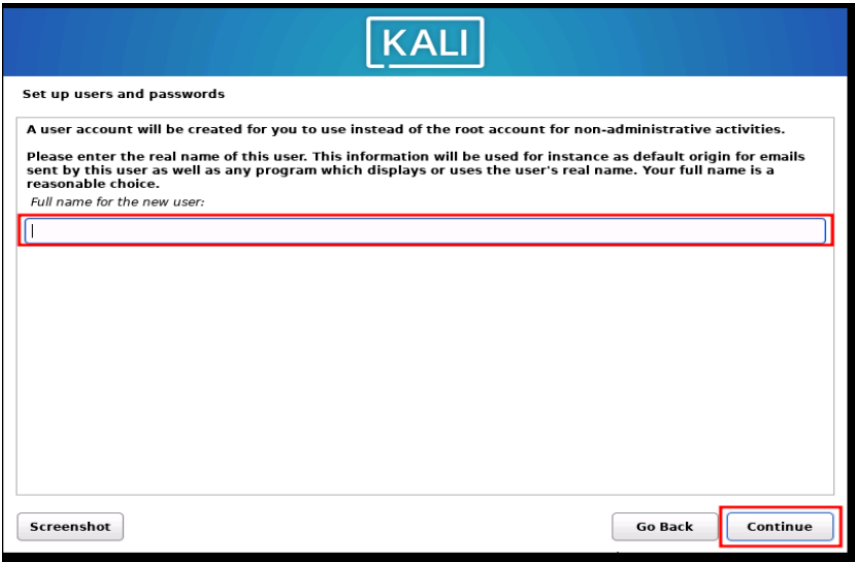
The screenshot shows the 'Configure the network' window in Kali Linux. At the top is a blue header with the 'KALI' logo. Below the header, the text 'Configure the network' is displayed. A message states: 'Please enter the hostname for this system. The hostname is a single word that identifies your system to the network. If you don't know what your hostname should be, consult your network administrator. If you are setting up your own home network, you can make something up here.' Below this message is a text input field labeled 'Hostname:' containing the text 'kali'. A red rectangle highlights this input field. At the bottom of the window, there are three buttons: 'Screenshot', 'Go Back', and 'Continue'. The 'Continue' button is highlighted with a red rectangle.

23. Leave the Domain name empty and continue.



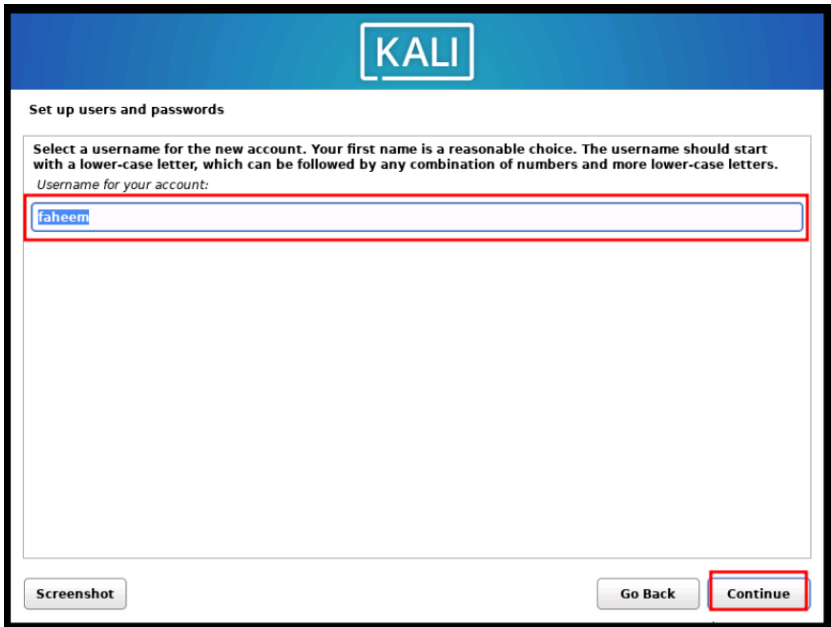
The screenshot shows the 'Configure the network' window in Kali Linux. At the top is a blue header with the 'KALI' logo. Below the header, the text 'Configure the network' is displayed. A message states: 'The domain name is the part of your Internet address to the right of your host name. It is often something that ends in .com, .net, .edu, or .org. If you are setting up a home network, you can make something up, but make sure you use the same domain name on all your computers.' Below this message is a text input field labeled 'Domain name:' which is empty. At the bottom of the window, there are three buttons: 'Screenshot', 'Go Back', and 'Continue'. The 'Continue' button is highlighted with a red rectangle.

24. Enter your full name or any name.



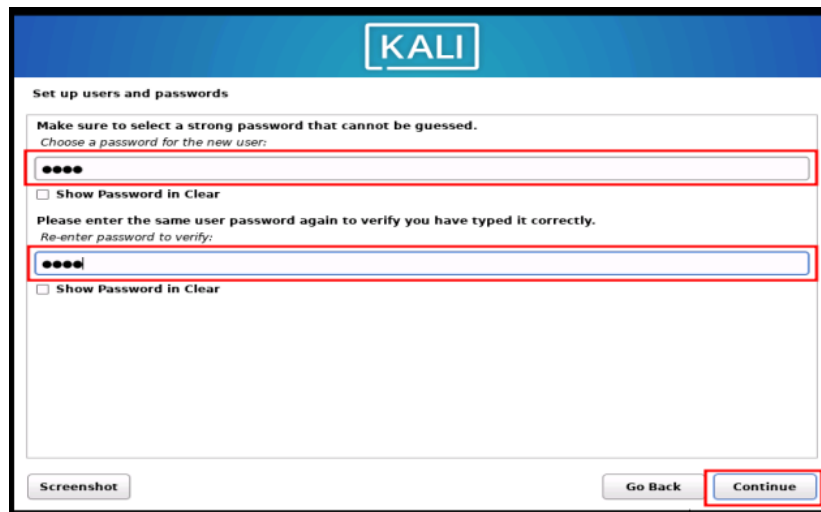
The screenshot shows the 'Set up users and passwords' screen in the KALI Linux installer. The title bar is blue with the 'KALI' logo. Below the title, the text reads: 'Set up users and passwords'. A message states: 'A user account will be created for you to use instead of the root account for non-administrative activities. Please enter the real name of this user. This information will be used for instance as default origin for emails sent by this user as well as any program which displays or uses the user's real name. Your full name is a reasonable choice.' Below this, a label says 'Full name for the new user:' followed by a large text input field. At the bottom, there are three buttons: 'Screenshot', 'Go Back', and 'Continue'. The 'Continue' button is highlighted with a red rectangle.

25. Set a username and click continue.



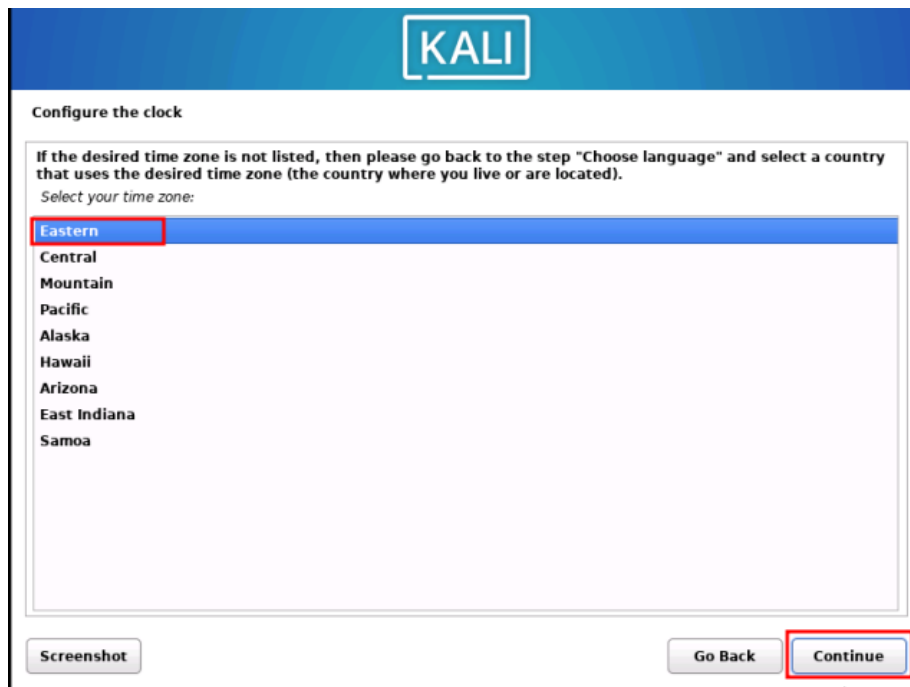
The screenshot shows the 'Set up users and passwords' screen in the KALI Linux installer. The title bar is blue with the 'KALI' logo. Below the title, the text reads: 'Set up users and passwords'. A message states: 'Select a username for the new account. Your first name is a reasonable choice. The username should start with a lower-case letter, which can be followed by any combination of numbers and more lower-case letters.' Below this, a label says 'Username for your account:' followed by a text input field containing the text 'faheem'. At the bottom, there are three buttons: 'Screenshot', 'Go Back', and 'Continue'. The 'Continue' button is highlighted with a red rectangle.

26. Set a password for your account.



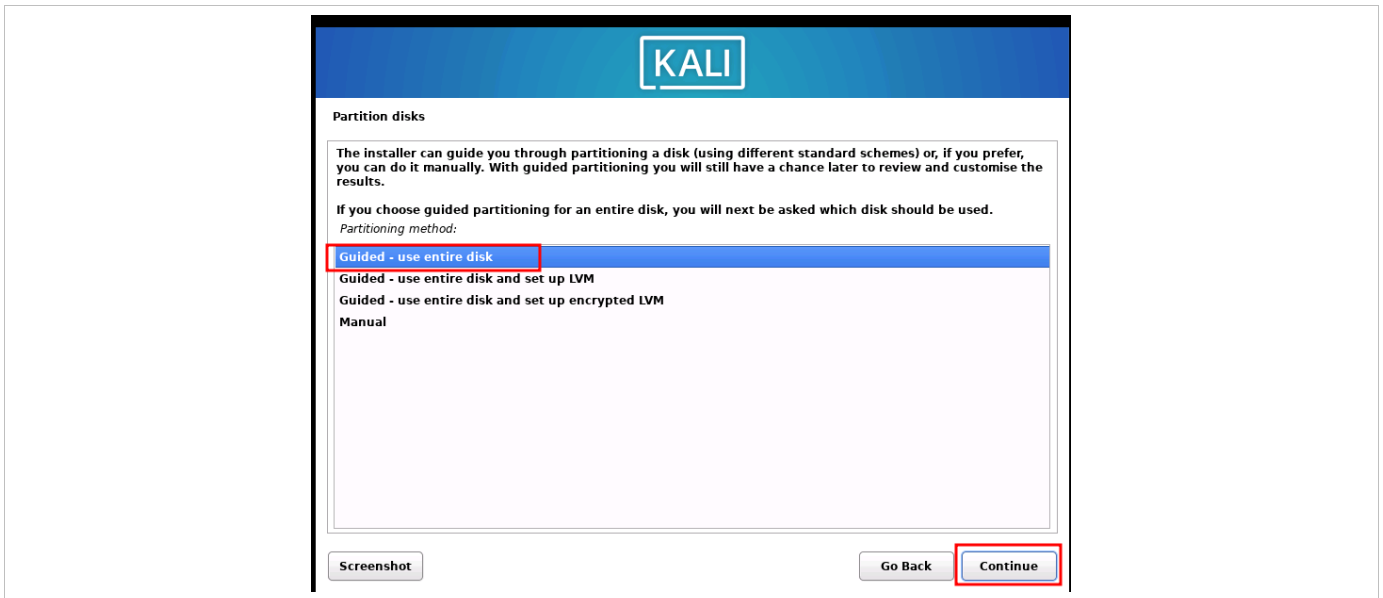
The screenshot shows the 'Set up users and passwords' screen in the KALI installer. The header has the 'KALI' logo. The main text says 'Set up users and passwords' and 'Make sure to select a strong password that cannot be guessed. Choose a password for the new user:'. Below this is a password input field with four dots, which is highlighted with a red box. There is a checkbox labeled 'Show Password in Clear'. Below that, it says 'Please enter the same user password again to verify you have typed it correctly. Re-enter password to verify:'. This is followed by another password input field with four dots, also highlighted with a red box. There is another checkbox labeled 'Show Password in Clear'. At the bottom, there are three buttons: 'Screenshot', 'Go Back', and 'Continue'. The 'Continue' button is highlighted with a red box.

27. Select the clock and continue.



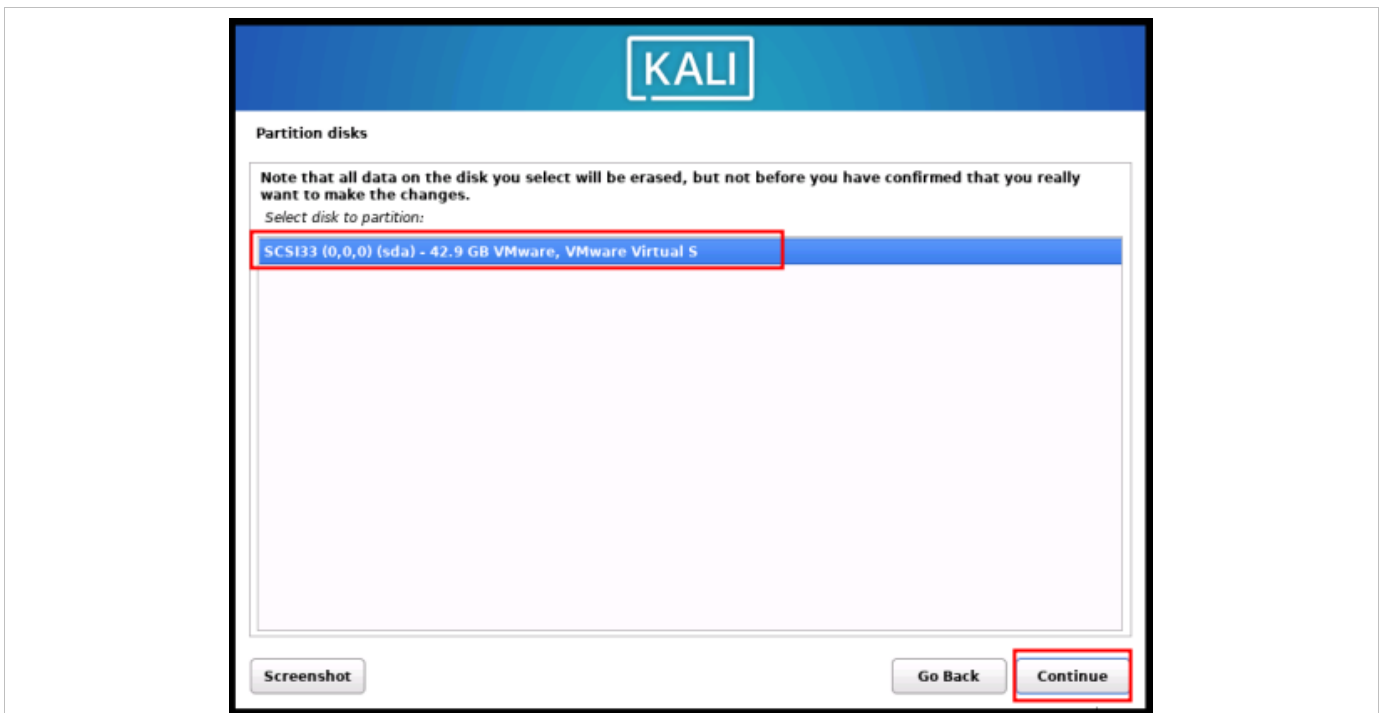
The screenshot shows the 'Configure the clock' screen in the KALI installer. The header has the 'KALI' logo. The main text says 'Configure the clock' and 'If the desired time zone is not listed, then please go back to the step "Choose language" and select a country that uses the desired time zone (the country where you live or are located). Select your time zone:'. Below this is a list of time zones: Eastern, Central, Mountain, Pacific, Alaska, Hawaii, Arizona, East Indiana, and Samoa. The 'Eastern' option is highlighted with a red box. At the bottom, there are three buttons: 'Screenshot', 'Go Back', and 'Continue'. The 'Continue' button is highlighted with a red box.

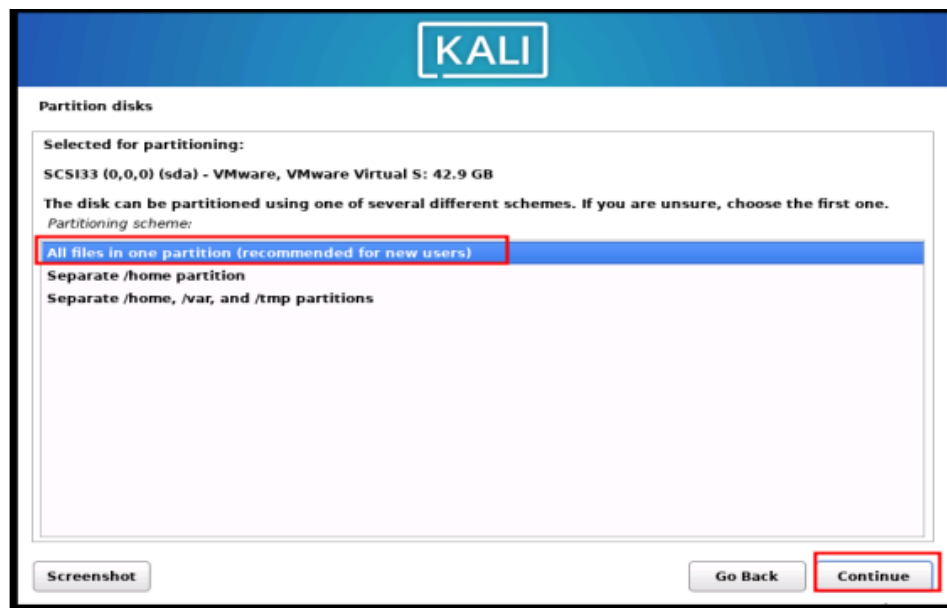
28. Choose Guided – use entire disk for partitioning, followed by All files in one partition.



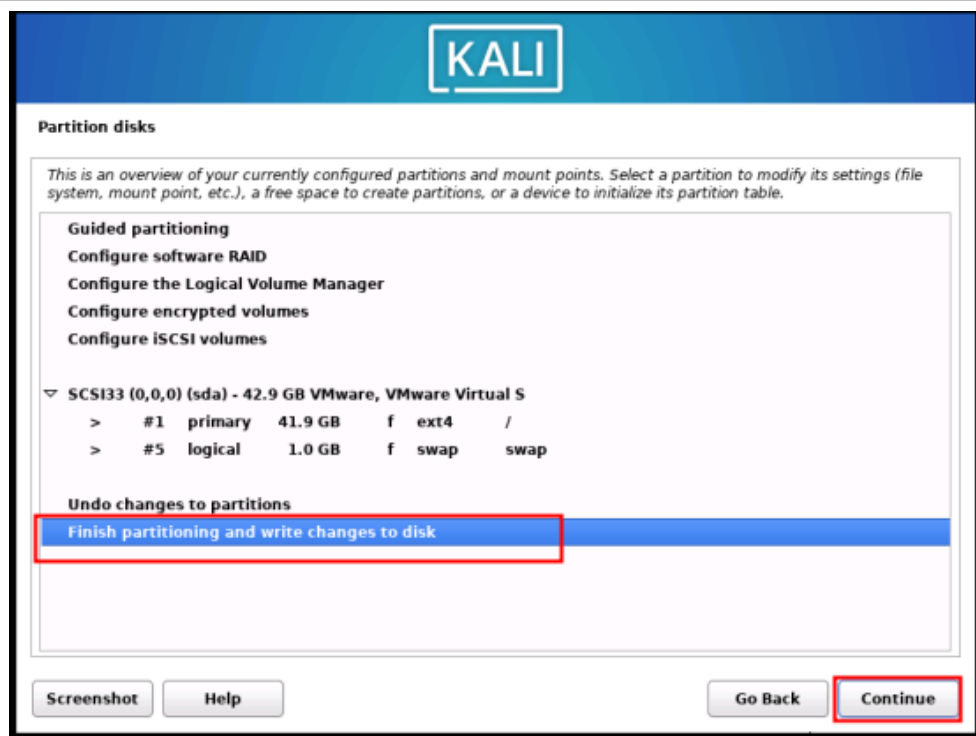
Guided partitioning; using the entire disk

29. Disk will be selected by default and click continue.

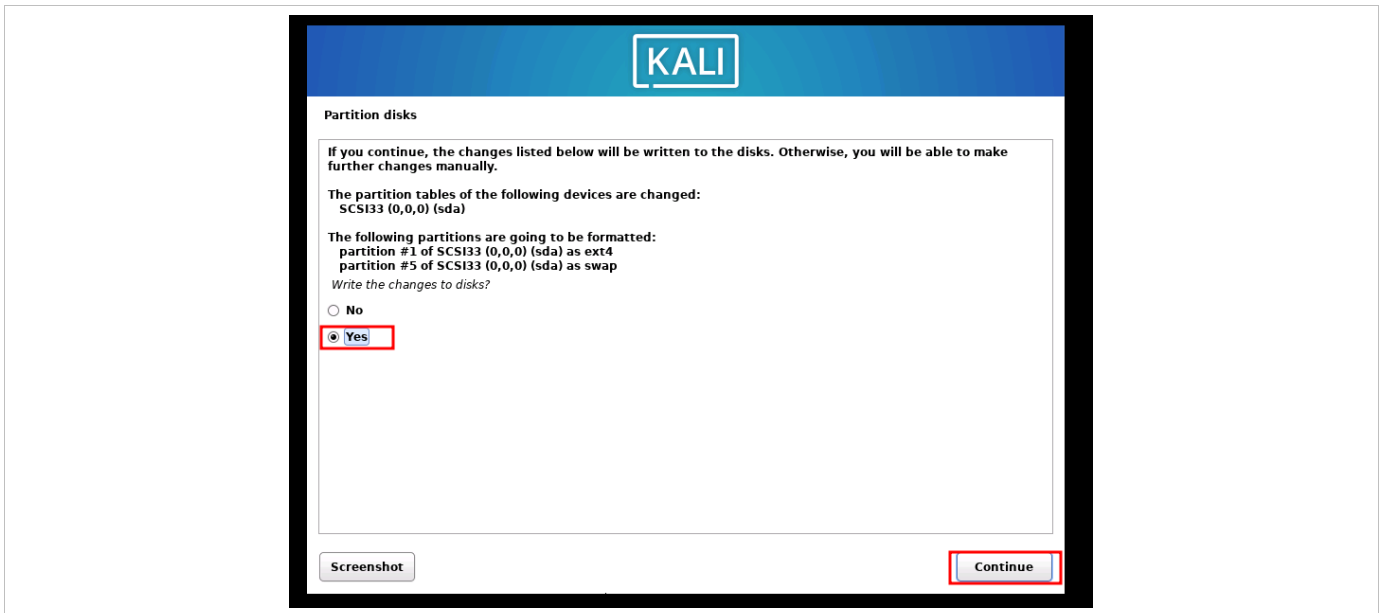




30. Verify disk partition then select Finish partitioning and click continue.

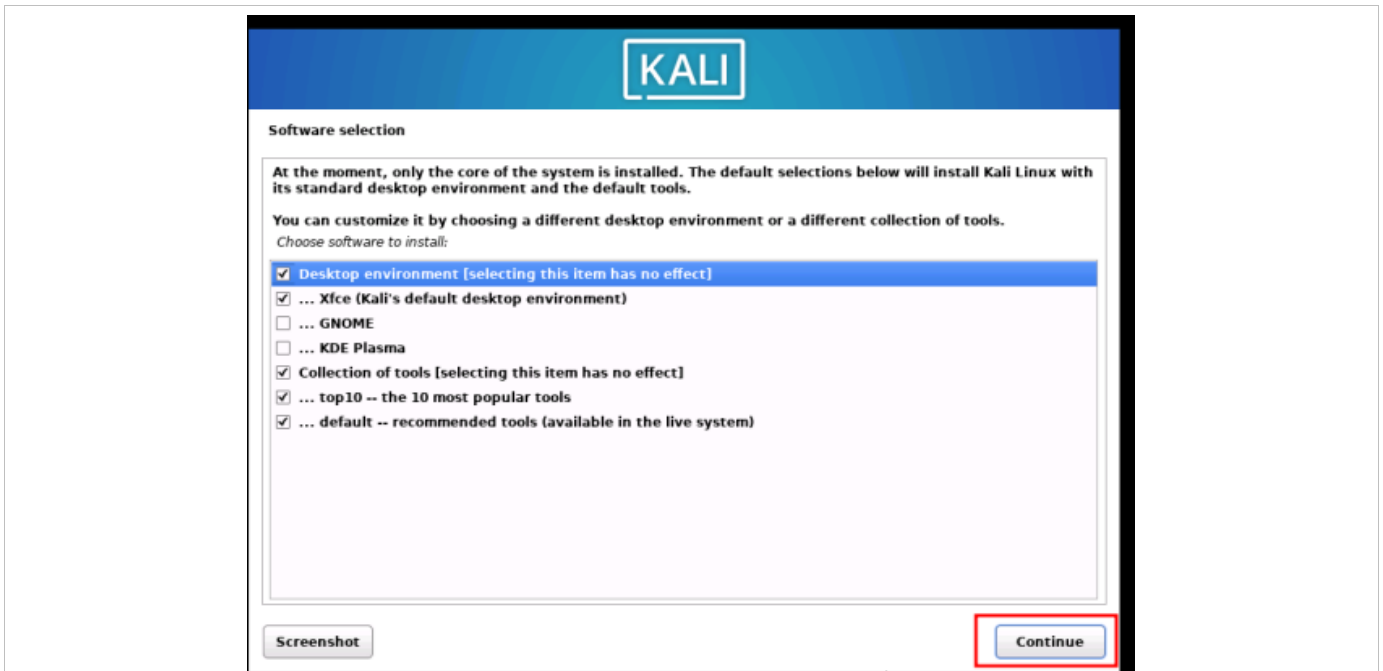


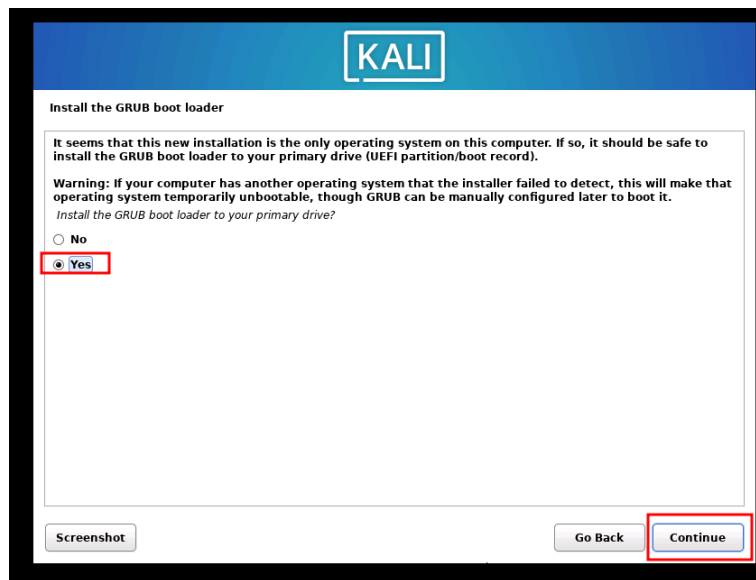
31. Confirm with Yes, and allow the installation to complete.



Confirming partition changes before installation

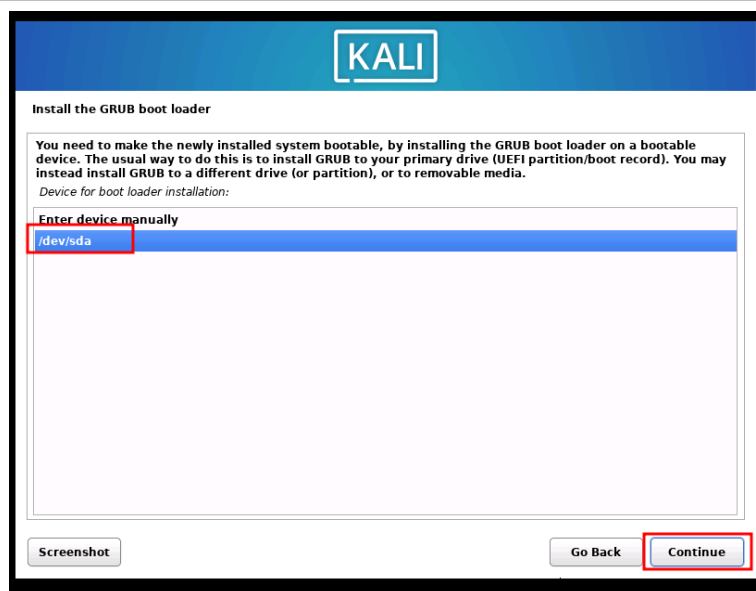
32. This will take sometime to install the complete operating system. When prompted, select a desktop environment (Xfce is Kali's default) and the default tool collection, then continue.





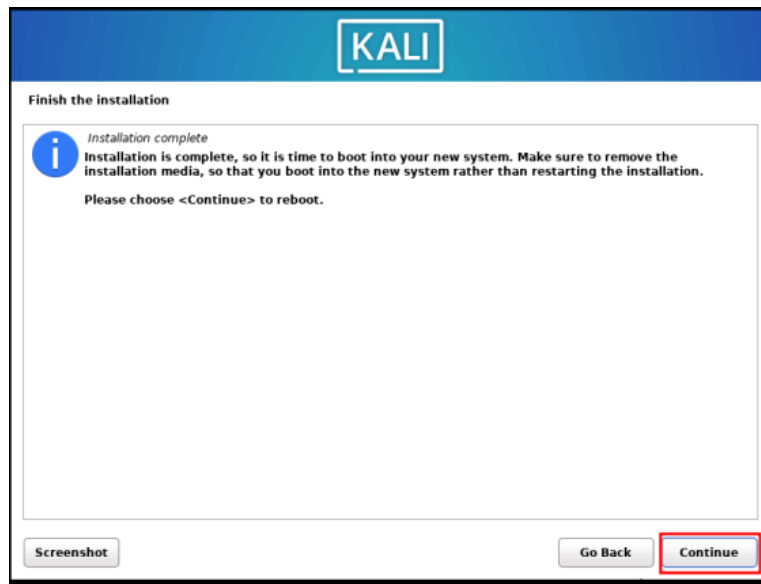
Software selection — desktop environment and tools

33. Install the GRUB bootloader to the primary drive (/dev/sda) when prompted, and click Continue to reboot.



Installing the GRUB bootloader

34. Click to continue to reboot the system, after rebooting the Kali Linux operating system will be installed and ready to use.



35. Log in using the username and password you created during installation.



Kali Linux login screen after reboot

Kali Linux Desktop Overview

Once logged in, note the following interface elements (referenced by number in your report screenshot):



Kali Linux Xfce desktop with key interface elements labeled

1. **Lock Screen Button:** Locks the screen or performs other power operations.
2. **Volume & Network Status Control:** Adjusts system volume and shows network, battery, and time status.
3. **Resource Monitor:** Displays live CPU and memory usage.
4. **Workspace Switcher:** Switches between virtual desktops.
5. **Terminal Icon:** Opens the command-line terminal.
6. **Show Desktop:** Minimizes/restores all open windows.
7. **Show Applications Icon:** Opens the applications menu.

Part B: Linux Fundamentals & Command-Line Tools (140 Marks)

Linux is a powerful, open-source operating system valued for its stability, security, and flexibility — it underpins most servers, embedded systems, and security tooling. In this part, you will practice essential command-line operations and file-management skills using Kali Linux.

Rules for Part B

- Download the task files from the link below before starting.
- Read each command's description in the Command Reference section before attempting the related task.
- Execute every required operation yourself — do not simply copy sample output shown in this document.
- For every task, record the exact commands you ran (in order) and include a full-screen terminal screenshot as evidence in your report.

Task Files Download Link:

<https://drive.google.com/file/d/1RFyK3X7Et1ZNfsMaDC4yEUcVn0N2CDIT/view?usp=sharing>

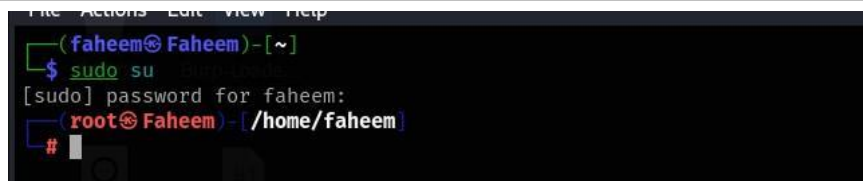
Command Reference

The commands below are used throughout the tasks in this section. Refer to this reference as needed; it will not be repeated elsewhere in the document.

SUDO Command

Short for “superuser do.” Lets you run a single command with administrative (root) privileges.

```
sudo su      # switch to the root user; the prompt changes from $ to #
```



Switching to the root user with sudo su

APT Command

Manages software packages on Debian-based systems (install, update, upgrade, remove). Typically requires sudo for system-wide changes.

```
apt update
apt upgrade
apt install <package>
```

```

root@Faheem:~/home/faheem
# apt install tree
tree is already the newest version (2.1.3-1).
tree set to manually installed.
The following packages were automatically installed and are no longer required:
 libravie0 libroc0.3 libsvtaviencid1 python3-diskcache python3-pendulum python3-pytzdata
Use 'sudo apt autoremove' to remove them.

Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0

root@Faheem:~/home/faheem
# apt update
Get:1 http://kali.download/kali kali-rolling InRelease [41.5 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [19.9 MB]
21% [2 Packages 7088 kB/19.9 MB 36%]

```

Installing a package and updating package lists with apt

MAN COMMAND

Displays the manual page for a given command, including all available options.

```
man cd
```

LS Command

Lists files and directories.

```

ls          # list contents
ls -l       # detailed listing (permissions, owner, size)
ls -a       # include hidden files
ls -la      # detailed listing, including hidden files

```

```

(faheem@Faheem)~/Desktop
$ ls
Backup  Burp-Loader-Linux-main  BurpSuite.sh  BurpSuite.zip  shellshocker

(faheem@Faheem)~/Desktop
$ ls -l
total 257448
drwxrwxrwx 5 faheem faheem 4096 Jun 25 01:18 Backup
drwxr-xr-x 2 faheem faheem 4096 Mar 24 02:24 Burp-Loader-Linux-main
-rwxr-xr-x 1 faheem faheem 181 Jul 8 04:29 BurpSuite.sh
-rw-rw-r-- 1 faheem faheem 263605859 Jul 22 13:51 BurpSuite.zip
drwxr-xr-x 6 faheem faheem 4096 Mar 25 06:06 shellshocker

(faheem@Faheem)~/Desktop
$ ls -la
total 257456
drwxr-xr-x 5 faheem faheem 4096 Aug 19 11:20 .
drwx----- 28 faheem faheem 4096 Aug 22 12:26 ..
drwxrwxrwx 5 faheem faheem 4096 Jun 25 01:18 Backup
drwxr-xr-x 2 faheem faheem 4096 Mar 24 02:24 Burp-Loader-Linux-main
-rwxr-xr-x 1 faheem faheem 181 Jul 8 04:29 BurpSuite.sh
-rw-rw-r-- 1 faheem faheem 263605859 Jul 22 13:51 BurpSuite.zip
drwxr-xr-x 6 faheem faheem 4096 Mar 25 06:06 shellshocker

(faheem@Faheem)~/Desktop
$ ls -a
.  ..  Backup  Burp-Loader-Linux-main  BurpSuite.sh  BurpSuite.zip  shellshocker

(faheem@Faheem)~/Desktop
$

```

Listing directory contents with ls, ls -l, ls -la

MKDIR Command

Creates a new directory.

```
mkdir project_folder
```



```
(faheem@Faheem) - [~/Desktop]
$ ls
Backup  Burp-Loader-Linux-main  BurpSuite.sh  BurpSuite.zip  shellshocker

(faheem@Faheem) - [~/Desktop]
$ mkdir i211910_MFaheem

(faheem@Faheem) - [~/Desktop]
$ ls
Backup  Burp-Loader-Linux-main  BurpSuite.sh  BurpSuite.zip  i211910_MFaheem  shellshocker

(faheem@Faheem) - [~/Desktop]
$
```

Creating a new directory with mkdir

PWD Command

Prints the full path of the current working directory.

```
pwd
```

```
(faheem@Faheem) - [~/Desktop]
$ pwd
/home/faheem/Desktop
```

Printing the current working directory with pwd

CD Command

Changes the current directory. Use ../ to move up one level.

```
cd project_folder
cd ../
```

```
(faheem@Faheem) - [~/Desktop]
$ ls
Backup  Burp-Loader-Linux-main  BurpSuite.sh  BurpSuite.zip  i211910_MFaheem  shellshocker

(faheem@Faheem) - [~/Desktop]
$ cd i211910_MFaheem

(faheem@Faheem) - [~/Desktop/i211910_MFaheem]
$
```

Changing directories with cd

TOUCH Command

Creates an empty file, or updates the timestamp of an existing one.

```
touch notes.txt
```

```
(faheem@Faheem) ~/Desktop/i211910_MFaheem
$ ls

(faheem@Faheem) ~/Desktop/i211910_MFaheem
$ touch i211910_c1.txt

(faheem@Faheem) ~/Desktop/i211910_MFaheem
$ ls
i211910_c1.txt

(faheem@Faheem) ~/Desktop/i211910_MFaheem
$
```

Creating an empty file with touch

CAT Command

Displays the contents of a file without opening an editor. Can also be combined with echo and redirection to write to a file.

```
cat notes.txt
echo "Hello World!" > notes.txt
```

```
(faheem@Faheem) ~/Desktop/i211910_MFaheem
$ ls
i211910_c1.txt

(faheem@Faheem) ~/Desktop/i211910_MFaheem
$ cat i211910_c1.txt

(faheem@Faheem) ~/Desktop/i211910_MFaheem
$ echo Hello World! > i211910_c1.txt

(faheem@Faheem) ~/Desktop/i211910_MFaheem
$ cat i211910_c1.txt
Hello World!

(faheem@Faheem) ~/Desktop/i211910_MFaheem
$
```

Displaying and writing file contents with cat and echo

CP Command

Copies a file or directory to a new location.

```
cp source.txt destination.txt
```

```

(faheem@Faheem)~/Desktop/i211910_MFaheem
$ ls
i211910_c1.txt

(faheem@Faheem)~/Desktop/i211910_MFaheem
$ cp i211910_c1.txt i211910_copy2.txt

(faheem@Faheem)~/Desktop/i211910_MFaheem
$ ls
i211910_c1.txt  i211910_copy2.txt

(faheem@Faheem)~/Desktop/i211910_MFaheem
$ cat i211910_c1.txt
Hello World!

(faheem@Faheem)~/Desktop/i211910_MFaheem
$ cat i211910_copy2.txt
Hello World!

(faheem@Faheem)~/Desktop/i211910_MFaheem
$

```

Copying a file with cp

MV Command

Moves or renames a file or directory.

```
mv source.txt target_folder/
```

```

(faheem@Faheem)~/Desktop/i211910_MFaheem
$ ls
i211910_c1.txt  i211910_copy2.txt

(faheem@Faheem)~/Desktop/i211910_MFaheem
$ mkdir test

(faheem@Faheem)~/Desktop/i211910_MFaheem
$ ls
i211910_c1.txt  i211910_copy2.txt  test

(faheem@Faheem)~/Desktop/i211910_MFaheem
$ ls test

(faheem@Faheem)~/Desktop/i211910_MFaheem
$ mv i211910_c1.txt test

(faheem@Faheem)~/Desktop/i211910_MFaheem
$ ls test
i211910_c1.txt

(faheem@Faheem)~/Desktop/i211910_MFaheem
$ ls
i211910_copy2.txt  test

(faheem@Faheem)~/Desktop/i211910_MFaheem
$

```

Moving a file into a directory with mv

RM/ RMDIR Command

rm removes files (use -r for directories and their contents). rmdir removes only empty directories.

```
rm file.txt
rm -r folder/
rmdir empty_folder/
```

A terminal window with a black background and green text. The prompt is '(faheem@Faheem)-[~/Desktop/i211910_MFaheem]'. The user enters '\$ ls' and the output is 'i211910_copy2.txt test'. Then the user enters '\$ rm i211910_copy2.txt'. The prompt changes to '(faheem@Faheem)-[~/Desktop/i211910_MFaheem]'. The user enters '\$ ls' and the output is 'test'. Finally, the user enters '\$' and the prompt changes to '(faheem@Faheem)-[~/Desktop/i211910_MFaheem]'.

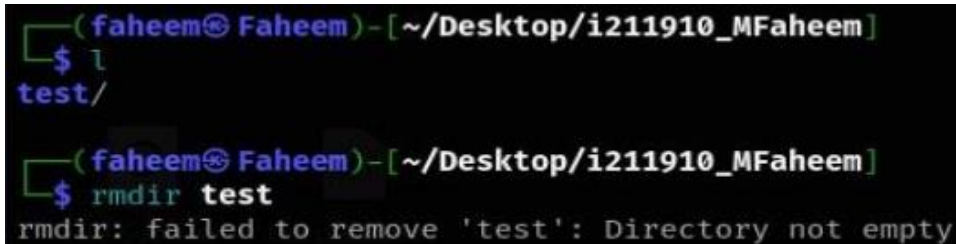
```
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ ls
i211910_copy2.txt  test

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ rm i211910_copy2.txt

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ ls
test

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$
```

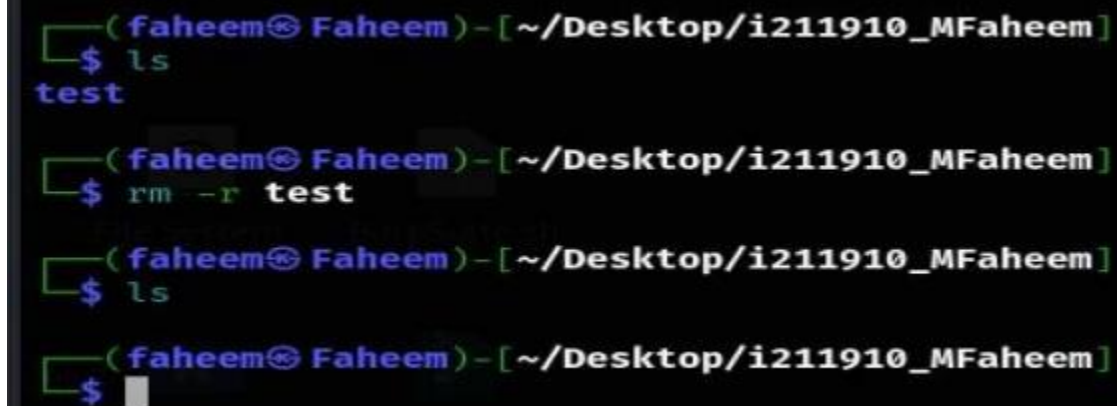
Removing a file with rm

A terminal window with a black background and green text. The prompt is '(faheem@Faheem)-[~/Desktop/i211910_MFaheem]'. The user enters '\$ ls' and the output is 'test/'. Then the user enters '\$ rmdir test'. The output is 'rmdir: failed to remove 'test': Directory not empty'.

```
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ ls
test/

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ rmdir test
rmdir: failed to remove 'test': Directory not empty
```

Removing an empty directories with rmdir

A terminal window with a black background and green text. The prompt is '(faheem@Faheem)-[~/Desktop/i211910_MFaheem]'. The user enters '\$ ls' and the output is 'test'. Then the user enters '\$ rm -r test'. The prompt changes to '(faheem@Faheem)-[~/Desktop/i211910_MFaheem]'. The user enters '\$ ls' and the output is empty. Finally, the user enters '\$' and the prompt changes to '(faheem@Faheem)-[~/Desktop/i211910_MFaheem]'.

```
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ ls
test

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ rm -r test

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ ls

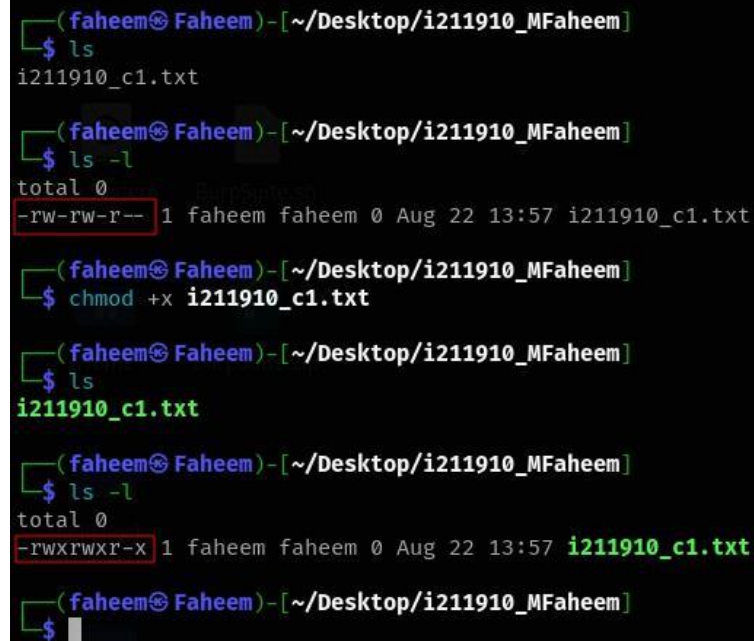
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$
```

Removing a non-empty directory with rm -r

CHMOD Command

Changes file or directory permissions. Symbolic mode (e.g., `chmod u+x file`) or numeric mode (e.g., `chmod 755 file`) may be used. Each numeric digit sums read (4), write (2), and execute (1) for owner, group, and others respectively. Add `-R` to apply recursively.

```
chmod u+x script.sh
chmod 700 secret.txt
```

A terminal window with a black background and green text. The prompt is '(faheem@Faheem)-[~/Desktop/i211910_MFaheem]'. The user runs 'ls', showing 'i211910_c1.txt'. Then they run 'ls -l', showing 'total 0' and '-rw-rw-r-- 1 faheem faheem 0 Aug 22 13:57 i211910_c1.txt'. The permissions '-rw-rw-r--' are highlighted with a red box. Next, they run 'chmod +x i211910_c1.txt'. Then they run 'ls', showing 'i211910_c1.txt' in green. Finally, they run 'ls -l', showing 'total 0' and '-rwxrwxr-x 1 faheem faheem 0 Aug 22 13:57 i211910_c1.txt'. The new permissions '-rwxrwxr-x' are highlighted with a red box. The prompt is now '(faheem@Faheem)-[~/Desktop/i211910_MFaheem]' with a cursor on the next line.

```
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ ls
i211910_c1.txt

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ ls -l
total 0
-rw-rw-r-- 1 faheem faheem 0 Aug 22 13:57 i211910_c1.txt

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ chmod +x i211910_c1.txt

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ ls
i211910_c1.txt

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ ls -l
total 0
-rwxrwxr-x 1 faheem faheem 0 Aug 22 13:57 i211910_c1.txt

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$
```

Changing file permissions with chmod

LOCATE Command

Quickly searches a prebuilt index for files or directories matching a name.

```
locate wordlist
```

```

(faheem@Faheem)-[~]
$ locate i211910
/home/faheem/Desktop/i211910_MFaheem
/home/faheem/Desktop/i211910_MFaheem/i211910_c1.txt

(faheem@Faheem)-[~]
$ locate wordlist
/etc/theHarvester/wordlists
/etc/theHarvester/wordlists/dns-big.txt
/etc/theHarvester/wordlists/dns-names.txt
/etc/theHarvester/wordlists/dorks.txt
/etc/theHarvester/wordlists/general
/etc/theHarvester/wordlists/names_small.txt
/etc/theHarvester/wordlists/general/common.txt
/home/faheem/.cache/dirb/resume/wordlist.dump
/usr/bin/wordlists
/usr/lib/python3/dist-packages/mnemonic/wordlist
/usr/lib/python3/dist-packages/mnemonic/wordlist/chinese_simplified.txt
/usr/lib/python3/dist-packages/mnemonic/wordlist/chinese_traditional.txt
/usr/lib/python3/dist-packages/mnemonic/wordlist/english.txt
/usr/lib/python3/dist-packages/mnemonic/wordlist/french.txt
/usr/lib/python3/dist-packages/mnemonic/wordlist/italian.txt
/usr/lib/python3/dist-packages/mnemonic/wordlist/japanese.txt
/usr/lib/python3/dist-packages/mnemonic/wordlist/korean.txt
/usr/lib/python3/dist-packages/mnemonic/wordlist/spanish.txt

```

Searching the filesystem index with locate

FIND Command

Searches for files and directories under a given path, with options to filter by name, type, or emptiness.

```

find /home/user -name "file.txt"
find . -type d          # directories only
find . -type f          # files only
find . -name "*.txt"    # by extension
find . -empty           # empty files/directories

```

```

(faheem@Faheem)-[~]
$ find /home/faheem/Desktop -name i211910_c1.txt
/home/faheem/Desktop/i211910_MFaheem/i211910_c1.txt

(faheem@Faheem)-[~]
$ find /home/faheem/Desktop/i211910_MFaheem/ -name i21*
/home/faheem/Desktop/i211910_MFaheem/
/home/faheem/Desktop/i211910_MFaheem/i211910_c1.txt

(faheem@Faheem)-[~]
$ find / -name i21*.txt 2> /dev/null
/home/faheem/Desktop/i211910_MFaheem/i211910_c1.txt
/home/faheem/Desktop/burpsuite.zip

(faheem@Faheem)-[~]
$

```

Searching for files by name with find


```

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ ls
Procfile  a.sh  i211910_c1.txt  shellshocker.py  test0  test1  test2

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ cd ~

(faheem@Faheem)-[~]
$ find /home/faheem/Desktop/i211910_MFaheem/ -type f
/home/faheem/Desktop/i211910_MFaheem/shellshocker.py
/home/faheem/Desktop/i211910_MFaheem/a.sh
/home/faheem/Desktop/i211910_MFaheem/i211910_c1.txt
/home/faheem/Desktop/i211910_MFaheem/Procfile

```

Searching for files by Type

```

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ ls
i211910_c1.txt  test0  test1  test2

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ cd ~

(faheem@Faheem)-[~]
$ find /home/faheem/Desktop/i211910_MFaheem/ -type d
/home/faheem/Desktop/i211910_MFaheem/
/home/faheem/Desktop/i211910_MFaheem/test0
/home/faheem/Desktop/i211910_MFaheem/test2
/home/faheem/Desktop/i211910_MFaheem/test1

(faheem@Faheem)-[~]
$

```

Searching for Directories

```

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ find /home/faheem/Desktop/i211910_MFaheem/ -name "*.txt"
/home/faheem/Desktop/i211910_MFaheem/i211910_c1.txt

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ find /home/faheem/Desktop/i211910_MFaheem/ -name "*.sh"
/home/faheem/Desktop/i211910_MFaheem/a.sh

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ find /home/faheem/Desktop/i211910_MFaheem/ -name "*"
/home/faheem/Desktop/i211910_MFaheem/
/home/faheem/Desktop/i211910_MFaheem/shellshocker.py
/home/faheem/Desktop/i211910_MFaheem/test0
/home/faheem/Desktop/i211910_MFaheem/a.sh
/home/faheem/Desktop/i211910_MFaheem/test2
/home/faheem/Desktop/i211910_MFaheem/i211910_c1.txt
/home/faheem/Desktop/i211910_MFaheem/test1
/home/faheem/Desktop/i211910_MFaheem/Procfile

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$

```

Searching for files by extension with find

```

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ find /home/faheem/Desktop/i211910_MFaheem/ -empty
/home/faheem/Desktop/i211910_MFaheem/test0
/home/faheem/Desktop/i211910_MFaheem/test2
/home/faheem/Desktop/i211910_MFaheem/i211910_c1.txt
/home/faheem/Desktop/i211910_MFaheem/test1

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ ls
Procfile  a.sh  i211910_c1.txt  shellshocker.py  test0  test1  test2

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$

```

Searching for empty files and directories

GREP Command

Searches text using patterns and prints matching lines. Use -i for case-insensitive search.

```

grep "password" file.txt
grep -i "username" file.txt

```

```

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ ls
Procfile  a.sh  grep_task.txt  i211910_c1.txt  shellshocker.py  test0  test1  test2  tmp.txt

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ grep "delete" i211910_c1.txt
- Remember to delete the messages from the phone.

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ grep "delete" i211910_c1.txt tmp.txt
i211910_c1.txt:- Remember to delete the messages from the phone.
tmp.txt:[12:32 PM] Alex: No way. We're in too deep. Just delete everything.

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ cat i211910_c1.txt | grep "delete"
- Remember to delete the messages from the phone.

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ grep "password" i211910_c1.txt

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ grep "pass" i211910_c1.txt

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ grep "Pass" i211910_c1.txt
- Password to the vault: 3sc@p3

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$

```

Searching file contents with grep

TR Command

Translates or deletes characters from input text — commonly used for case conversion, character removal, or substitution.

```

tr 'a-z' 'A-Z' < file.txt      # lowercase to uppercase
tr -s 'a-zA-Z'                # squeeze repeated characters
tr -d 'aeiouAEIOU' < file.txt # delete vowels

```



```
(faheem@Faheem)~[~/Desktop/i211910_MFaheem]
$ cat i211910_c1.txt
- Meeting with Alex at 3 PM. We need to discuss what happened.
- Remember to delete the messages from the phone.
- Check if the security camera footage was erased.
- Password to the vault: 3sc@p3
- She knows too much. Consider other options.
- Meeting place: The old warehouse at the docks.
- 8 PM tonight, bring the package.
- "It's not just about the money anymore."
- No loose ends. Ensure everything is taken care of.
- The last conversation with Alex was strange. Something is off.

(faheem@Faheem)~[~/Desktop/i211910_MFaheem]
$ cat i211910_c1.txt | tr -s '[a-z]' '[A-Z]'
- METING WITH ALEX AT 3 PM. WE NED TO DISCUS WHAT HAPENED.
- REMEMBER TO DELETE THE MESAGES FROM THE PHONE.
- CHECK IF THE SECURITY CAMERA FOTAGE WAS ERASED.
- PASWORD TO THE VAULT: 3SC@P3
- SHE KNOWS TO MUCH. CONSIDER OTHER OPTIONS.
- METING PLACE: THE OLD WAREHOUSE AT THE DOCKS.
- 8 PM TONIGHT, BRING THE PACKAGE.
- "IT'S NOT JUST ABOUT THE MONEY ANYMORE."
- NO LOSE ENDS. ENSURE EVERYTHING IS TAKEN CARE OF.
- THE LAST CONVERSATION WITH ALEX WAS STRANGE. SOMETHING IS OF.
```

Transforming text case with `tr`

HEAD Command

Displays first few lines of file.

[illegible]

Displaying the first lines of a file with head

TAIL Command

Displays last few lines of file.

[illegible]

Displaying the last lines of a file with tail

RESET Command

Resets the terminal to its default settings.

```
Erase is control-H (^H).
```

```
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]  
$
```

WC Command

Counts the number of lines, words, and characters in a file.

```
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]  
$ wc i211910_c1.txt  
11  90 490 i211910_c1.txt  
  
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]  
$ wc a.sh  
7  17 180 a.sh  
  
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]  
$ wc grep_task.txt  
176  176 29820 grep_task.txt  
  
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]  
$
```

For i211G10_c1.txt: 11 refers to the number of lines in the file. 90 refers to the number of words in the file.. 490 refers to the number of characters in the file

DU Command

Estimates and displays the disk space used by files and directories.

```
(faheem@Faheem)-[~/Desktop]  
$ du -h  
244K    ./Burp-Loader-Linux-main  
4.0K    ./i211910_MFaheem/test0  
4.0K    ./i211910_MFaheem/test2  
4.0K    ./i211910_MFaheem/test1  
68K     ./i211910_MFaheem  
20K     /shellshocker/shellshocker_server/templates  
  
(faheem@Faheem)-[~/Desktop]  
$ du -h /usr/share/wordlists  
139M    /usr/share/wordlists
```

Checking disk usage with du

AWK Command

A pattern-scanning and text-processing language. Supports variables such as NR (record number), FS (field separator), RS (record separator), and OFS (output field separator).

```
awk '{print}' file.txt
awk '{print $1, $4}' file.txt
awk 'BEGIN{FS=":"} {print $2}' file.txt
```

```
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ awk '{print}' awk_task.txt
ippsec youtube hackthebox 34024
john youtube ctf 50024
thecybermentor tcmsec courses 25923
liveoverflow youtube ctf 45345
nahamsec hackerone bughunting 12365
stok hackerone bughunting 1234
```

Printing file contents and fields with awk

```
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ awk '/ctf/' awk_task.txt
john youtube ctf 50024
liveoverflow youtube ctf 45345
```

Search for a pattern inside file

```
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ awk '{print $1 $4}' awk_task.txt
ippsec34024
john50024
thecybermentor25923
liveoverflow45345
nahamsec12365
stok1234
```

List me the words that are at 1st and 4th field

```
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ awk '{print NR,$0}' awk_task.txt
1 ippsec youtube hackthebox 34024
2 john youtube ctf 50024
3 thecybermentor tcmsec courses 25923
4 liveoverflow youtube ctf 45345
5 nahamsec hackerone bughunting 12365
6 stok hackerone bughunting 1234
```

NR: 'Number Record' is the variable that keeps count of the rows after each line's execution

```
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ awk 'BEGIN{FS="o"} {print $2}' awk_task.txt
youtube hacktheb
hn y
r tcmsec c
verfl
ne bughunting 12365
k hacker
```

FS: (Field Separator) is the variable to set in case you want to define the field for input stream

```
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ awk 'BEGIN{RS="o"} {print $0}' awk_task.txt
ippsec y
youtube hacktheb
x 34024
j
hn y
youtube ctf 50024
thecyberment
r tcmsec c
ourses 25923
live
```

RS: (Record Separator) By default, it separates rows with '\n', you can specify something else too.

```
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ awk '{print}' awk_task.txt
ippsec youtube hackthebox 34024
john youtube ctf 50024
thecybermentor tcmsec courses 25923
liveoverflow youtube ctf 45345
nahamsec hackerone bughunting 12365
stok hackerone bughunting 1234
```

Printing file contents and fields with awk

```
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ awk 'BEGIN{OFS=":"} {print $0}' awk_task.txt
ippsec youtube hackthebox 34024
john youtube ctf 50024
theycybermentor tcmsec courses 25923
liveoverflow youtube ctf 45345
nahamsec hackerone bughunting 12365
stok hackerone bughunting 1234
```

```
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ awk 'BEGIN{OFS=":"} {print $0,$1,$4}' awk_task.txt
ippsec youtube hackthebox 34024:ippsec:34024
john youtube ctf 50024:john:50024
theycybermentor tcmsec courses 25923:theycybermentor:25923
liveoverflow youtube ctf 45345:liveoverflow:45345
nahamsec hackerone bughunting 12365:nahamsec:12365
stok hackerone bughunting 1234:stok:1234
```

OFS: (Output Field Separator) Using OFS to format output with awk. You must have gathered some idea by the full form; it is to specify a delimiter while outputting

XARGS Command

Builds and executes commands using input received from standard input. It is useful for passing a list of items as positional arguments, especially alongside find.

```
find . -type f -print0 | xargs -0 rm -f
cat wordlist.txt | xargs -n 1 -t touch
```

```
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ echo "My name is Faheem aka MarkH0r" | xargs -t -I argVar sh -c '{ touch argVar; ls -l argVar; }'
sh -c '{ touch My name is Faheem aka MarkH0r; ls -l My name is Faheem aka MarkH0r; }'
-rw-rw-r-- 1 faheem faheem 0 Aug 24 02:00 Faheem
-rw-rw-r-- 1 faheem faheem 0 Aug 24 02:00 MarkH0r
-rw-rw-r-- 1 faheem faheem 0 Aug 24 02:00 My
-rw-rw-r-- 1 faheem faheem 0 Aug 24 02:00 aka
-rw-rw-r-- 1 faheem faheem 0 Aug 24 02:00 is
-rw-rw-r-- 1 faheem faheem 0 Aug 24 02:00 name

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ ls
Faheem Procfile awk_task.txt i211910_c1.zip shellshocker.py test2 xxd_task.txt
MarkH0r a.sh grep_task.txt is test0 tmp.txt
My aka i211910_c1.txt.gz name test1 xxd.txt
```

Building and running commands from input with xargs

```
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ ls ./test0
MarkH0r My is name shellshocker.py tmp.txt xxd.txt xxd_task.txt

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ find ./test0 -type f -print0 | xargs -0 -t -p rm -rf
rm -rf ./test0/shellshocker.py ./test0/My ./test0/xd_task.txt ./test0/tmp.txt ./test0/name ./test0/is
./test0/xd.txt ./test0/MarkH0r?...Y
Enter Y to confirm

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ ls ./test0

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$
```

Use find piped into xargs -0 -t -p rm -rf to safely delete files — -t prints each command before running, -p asks for confirmation, so nothing gets removed without review.


```

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ tar -xvf xargs_task.tar

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ ls
Faheem Procfile awk_task.txt i211910_c1.zip shellshocker.py test2 xargs_task.tar
MarKh0r a.sh grep_task.txt is test0 tmp.txt xxd.txt
My aka i211910_c1.txt.gz name test1 xargs xxd_task.txt

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ ls xargs
flag.txt one two

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ head xargs/flag.txt; wc xargs/flag.txt
eMfm5ES3tZBwYMFaxbCYv56fwX3eCtJn
sWscbzQ8JlghCBcvI7nupl64tbILLUYL
8x5y0uPFLWApFLsm3JwbVA0a7E63UdP9
QA4TasEIDTIV5GGXRrtD0kmtW74Gsvzm
HGIBQPCFebvmjeRcdF0XNXLXNBxX0Yoe
h1bTBhyZwACbyopIfNCKgQLogwLNDZ4c
XFckYhk9HvmrpyqNtVWZw1Vx6BXCrgvE
arReN9OLg5b1gMA2ZJQJtYRm86U4tqwJ
ai2QuCy1Ner0vpEZDeP3pcI1voASQxjF
lCho2Exctjmiq7a7UTT82sNQzEIVEI5C
1000 1000 33000 xargs/flag.txt

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ find . -type f -print0 2>/dev/null | xargs -0 egrep '^r[a-zA-Z0-9]*0$'
./xargs/flag.txt:rAJr4MvzgvIMJ9nPHfy6wWvRbvUM5J90
grep: ./xargs_task.tar: binary file matches

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$

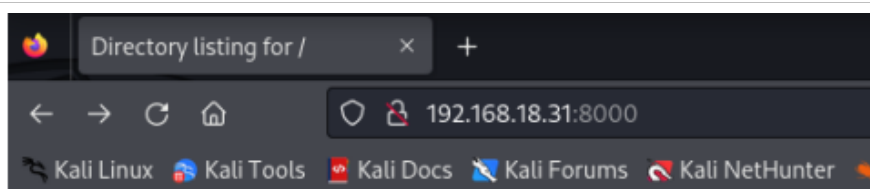
```

Extracted `xargs_task.tar`, then used `find . -type f -print0 | xargs -0 egrep '^r[a-zA-Z0-9]*0$'` to search every file for a string starting with `r` and ending in `0`. Found a unique match in `xargs/flag.txt`: `rAJr4MvzgvIMJ9nPHfy6wWvRbvUM5J90`.

CURL Command

Transfers data to or from a URL and prints the response. Supports custom headers, rate limiting, and uploads.

```
curl http://192.168.1.10:8000/
```



Directory listing for /

- [\[REDACTED\]](#)
- [awk_1611727975315.txt](#)
- [Compressed/](#)
- [desktop.ini](#)
- [Documents/](#)
- [FYP_1_Defense_SceneGen.pptx](#)
- [FYP_Registration_Form.docx](#)
- [FYP_Registration_Form.pdf](#)
- [grep_1611752025618.txt](#)
- [\[REDACTED\]](#)
- [\[REDACTED\]](#)
- [\[REDACTED\]](#)
- [\[REDACTED\]](#)
- [\[REDACTED\]](#)

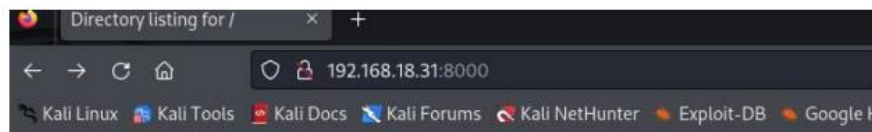
```
(faheem@faheem) - [~/Desktop/i211910_MFaheem]
$ curl http://192.168.18.31:8000/
<!DOCTYPE HTML>
<html lang="en">
<head>
<meta charset="utf-8">
<title>Directory listing for /</title>
</head>
<body>
<h1>Directory listing for /</h1>
<hr>
<ul>
<li><a href="FYP%201%20Defense_SceneGen.pptx">FYP_1_Defense_SceneGen.pptx</a></li>
<li><a href="awk_1611727975315.txt">awk_1611727975315.txt</a></li>
<li><a href="Compressed/">Compressed</a></li>
<li><a href="desktop.ini">desktop.ini</a></li>
<li><a href="Documents/">Documents</a></li>
<li><a href="FYP%201%20Defense_SceneGen.pptx">FYP_1_Defense_SceneGen.pptx</a></li>
<li><a href="FYP_Registration_Form.docx">FYP_Registration_Form.docx</a></li>
<li><a href="FYP_Registration_Form.pdf">FYP_Registration_Form.pdf</a></li>
<li><a href="grep_1611752025618.txt">grep_1611752025618.txt</a></li>
<li><a href="[REDACTED]">[REDACTED]</a></li>
<li><a href="[REDACTED]">[REDACTED]</li>
<li><a href="[REDACTED]">[REDACTED]</li>
</ul>
<hr>
</body>
</html>
```

Fetching a URL response with curl

WGET Command

Downloads files from the web directly via the command line.

```
wget http://192.168.1.10:8000/file.docx
```



Directory listing for /



```
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ wget http://192.168.18.31:8000/FYP_Registration_Form.docx
--2024-08-24 01:24:41-- http://192.168.18.31:8000/FYP_Registration_Form.docx
Connecting to 192.168.18.31:8000 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: 33934 (33K) [application/vnd.openxmlformats-officedocument.wordprocessingml.document]
Saving to: 'FYP_Registration_Form.docx'

FYP_Registration_Form.doc 100%[=====>] 33.14K --.-KB/s in 0s

2024-08-24 01:24:41 (108 MB/s) - 'FYP_Registration_Form.docx' saved [33934/33934]

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ ls
FYP_Registration_Form.docx  a.sh  grep_task.txt  i211910_c1.zip  test0  test2
Procfile                  awk_task.txt  i211910_c1.txt.gz  shellshocker.py  test1  tmp.txt
```

Downloading a file with wget

XXD Command

Produces a hex dump of a file (or reverses one). Useful for inspecting binary data, hex strings, and offsets.

```
xxd file.txt
xxd -s 0x10 file.txt # seek to byte 16
xxd -l 12 file.txt   # limit output length
```



```
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ echo "Hello World we are Fastians" | xxd
00000000: 4865 6c6c 6f20 576f 726c 6420 7765 2061 Hello World we a
00000010: 7265 2046 6173 7469 616e 730a re Fastians.

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ echo "Hello World we are Fastians" | xxd -E
00000000: 4865 6c6c 6f20 576f 726c 6420 7765 2061 ..%%? ..?..%...../
00000010: 7265 2046 6173 7469 616e 730a ...../ ... /> ..
```

(Use of -E Flag)

```
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ echo "Hello World we are Fastians" | xxd -b
00000000: 01001000 01100101 01101100 01101100 01101111 00100000 Hello
00000006: 01010111 01101111 01110010 01101100 01100100 00100000 World
0000000c: 01110111 01100101 00100000 01100001 01110010 01100101 we are
00000012: 00100000 01000110 01100001 01110011 01110100 01101001 Fasti
00000018: 01100001 01101110 01110011 00001010 ans.

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ echo "Hello World we are Fastians" | xxd -i
0x48, 0x65, 0x6c, 0x6c, 0x6f, 0x20, 0x57, 0x6f, 0x72, 0x6c, 0x64, 0x20,
0x77, 0x65, 0x20, 0x61, 0x72, 0x65, 0x20, 0x46, 0x61, 0x73, 0x74, 0x69,
0x61, 0x6e, 0x73, 0x0a
```

Output in binary and C include format

```
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ echo "Hello World we are Fastians" | xxd --l12
00000000: 4865 6c6c 6f20 576f 726c 6420 Hello World
```

Specifying a length

```
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ cat xxd.txt
hello
world
foo bar biz
my name is Faheem
i am Fastian
this is my last semester

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ xxd -s 0x10 xxd.txt
00000010: 6261 7220 6269 7a0a 6d79 206e 616d 6520 bar biz.my name
00000020: 6973 2046 6168 6565 6d0a 6920 616d 2046 is Faheem.i am F
00000030: 6173 7469 616e 0a74 6869 7320 6973 206d astian.this is m
00000040: 7920 6c61 7374 2073 656d 6573 7465 720a y last semester.
```

Seeking an Offset

```
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ xxd -s -16 xxd.txt
00000040: 7920 6c61 7374 2073 656d 6573 7465 720a y last semester.
```

Seeking an Offset from the end of the file

TREE Command

Displays a directory's contents as a tree-like structure.

```
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ tree
.
├── Procfile
├── a.sh
├── awk_task.txt
├── grep_task.txt
├── i211910_c1.txt
├── shellshocker.py
├── test0
├── test1
├── test2
└── tmp.txt

4 directories, 7 files
```

Displaying a directory structure with tree

FILE Command

Identifies a file's type based on its content rather than its extension.

```
file script.sh
```

```
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ file a.sh
a.sh: Bourne-Again shell script, ASCII text executable

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ file i211910_c1.txt
i211910_c1.txt: Unicode text, UTF-8 text

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ file shellshocker.py
shellshocker.py: Python script, ASCII text executable

(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$
```

Identifying file types with file

GZIP/ZIP Command

Compresses or decompresses files and directories.

```
zip archive.zip file.txt
gzip file.txt
```

```

(faheem@Faheem) ~/Desktop/i211910_MFaheem
$ ls
Procfile  awk_task.txt  i211910_c1.txt  test0  test2
a.sh      grep_task.txt shellshocker.py test1  tmp.txt

(faheem@Faheem) ~/Desktop/i211910_MFaheem
$ zip i211910_c1.zip i211910_c1.txt

adding: i211910_c1.txt (deflated 36%)

(faheem@Faheem) ~/Desktop/i211910_MFaheem
$ ls
Procfile  awk_task.txt  i211910_c1.txt  shellshocker.py  test1  tmp.txt
a.sh      grep_task.txt i211910_c1.zip  test0           test2

```

Compressing a file with zip

```

(faheem@Faheem) ~/Desktop/i211910_MFaheem
$ ls
Procfile  awk_task.txt  i211910_c1.txt  shellshocker.py  test1  tmp.txt
a.sh      grep_task.txt i211910_c1.zip  test0           test2

(faheem@Faheem) ~/Desktop/i211910_MFaheem
$ gzip i211910_c1.zip

(faheem@Faheem) ~/Desktop/i211910_MFaheem
$ ls
Procfile  awk_task.txt  i211910_c1.txt.gz  shellshocker.py  test1  tmp.txt
a.sh      grep_task.txt i211910_c1.zip    test0           test2

```

Compressing a file with gzip

NETSTAT Command

Displays active network connections, routing tables, and interface statistics.

```

(faheem@Faheem) ~/Desktop/i211910_MFaheem
$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp    0      0 192.168.85.128:33124    93.184.216.119:https    ESTABLISHED
udp    0      0 192.168.85.128:bootpc  192.168.85.254:bootps   ESTABLISHED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags   Type       State         I-Node  Path
unix  3      [ ]     STREAM    CONNECTED    17538
unix  3      [ ]     STREAM    CONNECTED    15131
unix  3      [ ]     SEQPACKET CONNECTED    501889
unix  3      [ ]     STREAM    CONNECTED    501881
unix  3      [ ]     STREAM    CONNECTED    8750      /run/systemd/journal/stdout
unix  3      [ ]     STREAM    CONNECTED    15199     @/tmp/.X11-unix/X0
unix  3      [ ]     STREAM    CONNECTED    9034
unix  2      [ ]     DGRAM     CONNECTED    9750
unix  3      [ ]     STREAM    CONNECTED    265797
unix  3      [ ]     STREAM    CONNECTED    17540
unix  2      [ ]     STREAM    CONNECTED    15117

```

Viewing active connections with netstat

IFCONFIG/IP A Command

Displays or configures network interface settings.

```
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether 02:42:7e:d9:63:d3 txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.85.128 netmask 255.255.255.0 broadcast 192.168.85.255
    inet6 fe80::56ea:d51e:a424:6f5 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:65:ce:24 txqueuelen 1000 (Ethernet)
    RX packets 56394 bytes 73667295 (70.2 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8343 bytes 614304 (599.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 157 bytes 12068 (11.7 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 157 bytes 12068 (11.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Viewing network interfaces with ifconfig

```
(faheem@Faheem)-[~/Desktop/i211910_MFaheem]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:65:ce:24 brd ff:ff:ff:ff:ff:ff
    inet 192.168.85.128/24 brd 192.168.85.255 scope global dynamic noprefixroute eth0
        valid_lft 1710sec preferred_lft 1710sec
    inet6 fe80::56ea:d51e:a424:6f5/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 02:42:7e:d9:63:d3 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
```

Viewing network interfaces with ip a

SYSTEMCTL Command

Manages system services — checking status, starting, stopping, restarting, enabling, or disabling them at boot.

```
systemctl status apache2
sudo systemctl start apache2
sudo systemctl enable apache2
```



```

(faheem@Faheem) - [~/Desktop/i211910_MFaheem]
$ sudo systemctl start apache2
[sudo] password for faheem:

(faheem@Faheem) - [~/Desktop/i211910_MFaheem]
$ systemctl status apache2

● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; disabled; preset: disabled)
   Active: active (running) since Sat 2024-08-24 02:41:16 EDT; 6s ago
 Invocation: c74a553766a24ecfa347e9f3ed2d7e25
    Docs: https://httpd.apache.org/docs/2.4/
 Process: 304308 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
 Main PID: 304332 (apache2)
   Tasks: 6 (limit: 9393)
  Memory: 19.6M (peak: 20.3M)
     CPU: 1.178s
   CGroup: /system.slice/apache2.service
           └─304332 /usr/sbin/apache2 -k start
             └─304335 /usr/sbin/apache2 -k start
               └─304336 /usr/sbin/apache2 -k start

```

Starting a service and checking its status with systemctl

```

(faheem@Faheem) - [~/Desktop/i211910_MFaheem]
$ sudo systemctl restart apache2

```

This command restarts the Apache2 service

```

(faheem@Faheem) - [~/Desktop/i211910_MFaheem]
$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-inst
all.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
Created symlink '/etc/systemd/system/multi-user.target.wants/apache2.service' -> '/usr/lib/systemd/syste
m/apache2.service'.

```

This command configures the Apache2 service to start automatically when the system boots.

```

(faheem@Faheem) - [~/Desktop/i211910_MFaheem]
$ sudo systemctl disable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-inst
all.
Executing: /usr/lib/systemd/systemd-sysv-install disable apache2
Removed '/etc/systemd/system/multi-user.target.wants/apache2.service'.

```

This command prevents the Apache2 service from starting automatically at boot time

```

(faheem@Faheem) - [~/Desktop/i211910_MFaheem]
$ systemctl list-units --type=service --state=active

```

UNIT	LOAD	ACTIVE	SUB	DESCRIPTION
accounts-daemon.service	loaded	active	running	Accounts Service
apache2.service	loaded	active	running	The Apache HTTP Server
binfmt-support.service	loaded	active	exited	Enable support for additional executables
colord.service	loaded	active	running	Manage, Install and Generate Color Profiles
console-setup.service	loaded	active	exited	Set console font and keymap
containerd.service	loaded	active	running	containerd container runtime
cron.service	loaded	active	running	Regular background program processing
dbus.service	loaded	active	running	D-Bus System Message Bus
docker.service	loaded	active	running	Docker Application Container Engine
getty@tty1.service	loaded	active	running	Getty on tty1
haveged.service	loaded	active	running	Entropy Daemon based on the HAVEGE algorithm
ifupdown-pre.service	loaded	active	exited	Helper to synchronize boot up for ifupdown
keyboard-setup.service	loaded	active	exited	Set the console keyboard layout
kmod-static-nodes.service	loaded	active	exited	Create List of Static Device Nodes
lightdm.service	loaded	active	running	Light Display Manager
ModemManager.service	loaded	active	running	Modem Manager
networking.service	loaded	active	exited	Raise network interfaces
NetworkManager-wait-online.service	loaded	active	exited	Network Manager Wait Online
NetworkManager.service	loaded	active	running	Network Manager

This command lists all currently active services on the system

Assessment Tasks

Use the roll-number placeholder i23xxxx consistently throughout Part B. Replace it with your own roll number in every file name, directory name, and script.

Task 1: Root Access with sudo (5 marks)

- Open your terminal.
- Run `sudo su` to switch to the root user.
- Confirm the switch by checking that the prompt changes to `#` (indicating root access).

Task 2: Package Management with apt (5 marks)

- Update and upgrade the system package lists.
- Install the `tree` and `wget` packages.
- Run the `tree` command and observe the output.
- Use `wget` to download any file from the internet.

Task 3: File & Directory Management (10 marks)

- Create a directory named `i23xxxx_Name` (replace with your own name and roll number).
- Navigate into the new directory.
- Confirm your current location.
- Create an empty file named `i23xxxx_c1.txt`.
- Add a brief introduction about yourself to the file.
- Display the contents of `i23xxxx_c1.txt`.
- Append the text "Project Started" to the file.
- Create a copy of the file named `backup_i23xxxx.txt`.
- Move the backup file to the parent directory.
- Navigate back to the parent directory and confirm the backup file exists.

Task 4: Searching & Scripting with grep and Python (10 marks)

- Using the provided `grep_task.txt`, run `cd /` and then use `locate` to find the file's path from root.
- Also use `find` to locate `grep_task.txt` starting from your current directory.
- Use `grep` (case-insensitive) to find the username mentioned in the file.
- Identify the password associated with that username.
- Search the file for any comment left by the user.
- Create a Python file named `i23xxxx_addition.py` that takes two numbers as input, adds them, and prints the result in the format: "Hey i23xxxx, the sum of Two Numbers is: ..."
- Change the file's permissions (numeric mode) so that only the owner can read, write, and execute it — no access for anyone else.
- Run `i23xxxx_addition.py` to confirm it still executes correctly under the new permissions.
- Use `ls -l` to display and confirm the updated permissions, and include a screenshot.

Task 5: Text Manipulation with tr (10 marks)

- Run `tr --help` and explain how to match any single digit character in a string.
- State the character-class sequence equivalent to `[a-zA-Z]`.
- State the character-class sequence equivalent to matching hexadecimal characters.
- Using the provided `grep_task.txt`, convert all lowercase letters to uppercase with `tr`.
- Remove all vowels (uppercase and lowercase) from the text using `tr`.
- Replace all spaces in the file with underscores using `tr`.
- Compress (squeeze) any sequences of repeated characters using `tr`.
- Delete all non-alphanumeric characters from the file using `tr`.
- Replace all tab characters in the file with spaces using `tr`.

Task 6: Text Processing with awk (10 marks)

- Using the provided `awk_task.txt`, use `awk` to print each entry in the format `name:score` (one pair per line, using the 1st and 4th fields).
- Explain — using the `BEGIN` pattern and the `FS/OFS/RS` variables covered in the Command Reference — at least one way to instead print all names as a single comma-separated line (e.g., `name1, name2, name3, ...`).

Task 7: Batch Operations with xargs (20 marks)

- Using the provided `xarg_task.txt` and the file list inside the "one" folder, use `xargs` to create every listed file in your current directory in a single command, then set their permissions to owner-read-only (numeric mode). Apply the `-t` (verbose) flag and pass the file list as the positional argument, using the flags in ASCII order.
- Using the file list(s) provided across the "one" and "two" folders, rebuild the shortened wordlist (shortrockyou) in a single `xargs` command line, limiting execution to one argument per command invocation. Apply the `-t` (verbose) flag together with the flag that limits arguments per line, using the flags in ASCII order.
- Which `xargs` flag limits the number of arguments passed to each command execution?
- How do you tell `xargs` to treat everything that follows as positional arguments rather than as flags (i.e., how do you escape flag-like input)?

Task 8: Transferring Data with curl (5 marks)

- Which `curl` flag limits the download or upload transfer rate of a file?
- Run: `curl -A 'juzztesting' https://nuces.tools/`
- Can `curl` perform file upload operations? (Yes / No)

Task 9: Downloading Files with wget (5 marks)

- How do you enable timestamp logging for every new activity that `wget` performs?
- Write the command to download `https://xyz.com/mypackage.zip` using `wget` while appending its log output to an existing file named `package-logs.txt`.
- Write the command to read a list of URLs from `file.txt` and limit the download speed to 1 Mbps.

Task 10: Hex Inspection with xxd (10 marks)

- Using the provided `xxd_task.txt`, seek to the 10th byte (in hex) and display only the next 50 bytes.

- Display the hex dump in 3 output columns, grouped in sets of 3 octets per row (use the relevant flags in alphabetical order).
- Which flag takes precedence when both -c and -g are supplied together?
- Locate and report the value of "flag" contained in xxd_task.txt.

Task 11: Managing Services with systemctl (20 marks)

- Install openssh-server via the terminal.
- Check whether the SSH service is currently running.
- If it is not running, start it and verify its status afterward.
- Stop the SSH service and confirm that it has stopped.
- Restart the SSH service and note any changes in its status.
- Configure the SSH service to start automatically on boot.
- State the command used to prevent the SSH service from starting automatically on boot.
- Edit the SSH configuration file (/etc/ssh/sshd_config) and reload the service to apply the change without a full restart.
- List all units related to SSH, including sockets.
- Intentionally introduce an error into the SSH configuration file, attempt to restart the service, use its status output to diagnose the failure, and then correct the error.

Task 12: Security Aspects of Your Final Year Project (FYP) (30 marks)

- Assets: Identify what needs protection in your FYP (e.g., data, source code, user accounts, devices).
- Threats & Risks: Identify the possible security risks or attacks that could affect your project.
- Vulnerabilities: Identify where weaknesses could appear (e.g., weak authentication, insecure data storage, unencrypted communication).
- Mitigation Strategies: Describe the technical, procedural, or design-level measures you would apply to reduce or prevent each identified threat.